

READING COMMODITY NEWS TO NAVIGATE FINANCIAL MARKETS: SUPPLY, DEMAND, AND THE REGIME-DEPENDENCE OF INFLATION RISK

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Abstract

We construct R_t , a monthly real-time indicator of whether commodity price dynamics are demand- or supply-driven, built from the relative volatility of demand and supply signals across over one million Reuters commodity news articles spanning 15 commodity categories. Using local projections, we show that R_t significantly predicts the rolling inflation-output gap correlation twelve months ahead ($\beta = 2.02$, $t = 3.61$), providing a leading real-time signal of the macro regime formalized by Campbell, Pflueger, and Viceira (2020). In demand-dominated months, R_t correctly identifies when bonds hedge equities in more than 70% of cases. In a conditional Fama-MacBeth framework following Fang, Liu, and Roussanov (2022), an energy-specific version of R_t resolves a documented open puzzle: all seven asset classes show larger energy inflation betas in demand-dominated months than in supply-dominated months, a 7/7 result robust across rolling windows of 12 to 60 months. Core inflation risk remains consistently and negatively priced regardless of regime. Together, our findings show that the origin of a commodity price movement recoverable in real time from the news predicts macroeconomic regimes, portfolio correlations and the cross-sectional pricing of inflation risk.

Keywords: commodity price shocks, demand and supply decomposition, monetary policy regimes, stock-bond correlation, local projections, inflation risk pricing

JEL Codes: E31, E52, G12, Q02

1. Introduction

When commodity prices rise sharply, the headline number is ambiguous. Consider two episodes: 2020–2021 and 2022. In 2020–2021, global commodity prices rose because the world was reopening: fiscal stimulus flooded households with income, industrial activity rebounded, and Chinese demand accelerated. In 2022, they rose because Russia invaded Ukraine, severing major supply sources at a moment when global demand was already softening from tightening financial conditions. The price increases in both cases were large and rapid. Their macroeconomic meanings

were opposite. Their portfolio consequences diverged sharply: in the first episode, bonds continued to hedge equities and the 60/40 portfolio held. In the second, stocks and bonds fell together and the hedge broke down. The same commodity price increase, two entirely different worlds.

This paper asks whether those consequences can be anticipated in real time, before official data arrives, by reading the news. The theoretical foundation for why this should be possible comes from Campbell, Pflueger, and Viceira (2020), hereafter CPV. Their central insight is that the covariance between inflation and real economic activity determines the hedging value of bonds. When demand shocks dominate, with the economy booming and inflation and employment rising together, the central bank can cut rates freely when equities fall, pushing bond prices up: bonds are safe havens. The classic 60/40 portfolio works. When supply shocks dominate and inflation rises as output weakens, the central bank faces a dilemma and can not simultaneously stabilize prices and support growth. It can't cut rates because inflation is already high. Bond prices can not rally to offset equity losses and the stock-bond correlation turns positive. Despite its theoretical clarity, the CPV framework is difficult to operationalize prospectively. The inflation-output correlation is a realized quantity, measurable only after official data is published. Identifying in real time whether the current environment is demand- or supply-dominated is considerably harder. Investors watching commodity prices spike in March 2022 could not know from the theory alone which regime they were entering.

A second open puzzle comes from Fang, Liu, and Roussanov (2022), hereafter FLR. They show that the price of core inflation risk is consistently and significantly negative across every asset class such as stocks, bonds, currencies, REITs and commodities because core inflation signals lower future real growth and lower dividends. The price of energy inflation risk, by contrast, is positive on average but statistically unstable across time. FLR explicitly attribute this instability to whether energy inflation reflects booming demand or a supply disruption, but they have no real-time instrument to test this hypothesis.

Both gaps point to the same missing object: a monthly, real-time signal that identifies whether commodity price movements are demand-driven or supply-driven across the full commodity universe, available before any official macro data arrives.

Our contribution is to show that a systematic reading of commodity news provides a high-frequency, forward-looking signal for exactly this purpose. We build that signal. We construct R_t , a monthly indicator of whether commodity news skews toward demand or supply language, using the text-based indices of Lumbanraja, Mouabbi, Passari, and Rousset Planat (2025), hereafter LMPRP, who apply a supervised natural language processing algorithm to more than 1,100,000 Reuters articles covering 15 commodity categories from May 2000 to May 2023. Malliaropoulos, Passari, and Petroulakis (2025), hereafter MPP, show that these indices contain distinct predictive information for inflation beyond commodity prices or survey forecasts. We define R_t as the ratio of demand-signal volatility to total commodity news volatility over a 24-month rolling window. High R_t signals a demand-dominated environment. Low R_t signals supply dominance. The indicator is real-time by construction: it uses only news published before any official macro release.

This paper is organized around three completed empirical exercises. The first concerns the macro regime. Using local projections following Jordà (2005), R_t significantly predicts the rolling inflation-output gap correlation at a 12-month horizon, with a coefficient that grows monotonically as the horizon extends. The indicator leads the macro regime rather than merely tracking it, and the result holds across all rolling window lengths we try and in every sub-period where monetary policy transmission is intact. The second concerns the stock-bond correlation. In demand-dominated months, bonds hedge equities in more than 70% of cases, well above the 50% coin-flip benchmark. In a horse race against established competitors, a contemporaneous realized energy volatility measure dominates at short horizons, but R_t recovers comparable predictive content at a 12-month horizon. This pattern is exactly what a leading indicator should produce: the news signal matters most at the horizon where backward-looking realized measures have not yet had time to reflect the regime shift. The third resolves an open puzzle in cross-sectional asset pricing. We build an energy-specific version of the indicator from the oil, natural gas, and gasoline sub-indices of LMPRP and use it to classify months into demand-dominated and supply-dominated regimes. Across all seven asset classes we study. Every single class shows a larger energy inflation beta in demand months than in supply months.

The paper proceeds as follows. Section 2 reviews the related literature. Section 3 describes the data and the construction of all variables. Section 4 presents the inflation-output regime results.

Section 5 presents the stock-bond correlation analysis. Section 6 presents the conditional pricing of inflation risk across asset classes. Section 7 concludes.

2. Related Literature

Identifying commodity shocks. The foundational challenge in commodity economics is that price movements are observationally ambiguous. A price rise could reflect booming global demand or a disruption to supply, and these two origins have opposite macroeconomic implications. Kilian (2009) addresses this for oil markets through a structural VAR, distinguishing oil supply shocks, aggregate demand shocks, and oil-specific demand shocks. Känzig (2021) sharpens identification using high-frequency variation around OPEC announcements. These structural approaches are powerful but narrow in scope. They require commodity-specific data, strong modelling assumptions, and can not easily extend beyond oil to metals, agriculture, or livestock. LMPRP (2025) and MPP (2025) broaden coverage to the full commodity spectrum by reading the news directly. Applying a supervised natural language processing algorithm to more than one million Reuters articles, they produce monthly supply and demand indicators for more than fifteen commodity categories simultaneously. A key empirical finding of LMPRP is that demand shocks travel across commodities while supply shocks are idiosyncratic to each market. MPP then show that these text-based indicators contain distinct predictive power for inflation beyond commodity price levels, survey forecasts, or conventional macroeconomic controls. We build on both papers. The signal we construct, R_t , translates the LMPRP indices into a monthly, real-time regime indicator available before any official macro release.

Macro regimes and the stock-bond correlation. Whether bonds hedge equities is not a fixed property of the two asset classes. It depends on the macroeconomic regime. CPV (2020) show that the sign of the stock-bond correlation is determined by the covariance between inflation and real activity. When demand shocks dominate, inflation and output move together. The central bank cuts rates freely when equities fall, bond prices rally, and bonds serve as safe havens. The stock-bond correlation is negative. When supply shocks dominate, inflation rises as output weakens. The central bank faces a dilemma it cannot resolve simultaneously: fighting inflation deepens the recession, while protecting growth lets inflation run. Bond prices cannot rally to offset equity losses, and the stock-bond correlation turns positive. CPV identify a structural break around

2001 using a Quandt likelihood ratio test: before 2001, the inflation-output correlation was negative and the stock-bond correlation was positive; after 2001, both reversed. The model fits both periods with the same preference parameters. But CPV identify the regime only ex post, using realised quarterly correlations over long historical windows. Investors can not observe the inflation-output correlation for the current quarter. This is the first gap our paper addresses.

Bekaert and Engstrom (2010) corroborate the CPV mechanism from a different angle. They show that the positive correlation between equity yields and bond yields (the so-called Fed model) arises rationally through time-varying risk premia rather than money illusion. Their cross-country evidence is compelling: countries with a higher historical incidence of stagflation display significantly higher equity-bond yield co-movement. What matters is the simultaneous occurrence of inflation with weak output.

Cieslak and Pang (2021) bring high-frequency identification to the same question. They decompose daily stock returns and Treasury yield changes into four structural shocks using a sign-restricted VAR estimated from 1983 to 2017. Around 80 percent of variation in 10-year Treasury yields comes from risk premium shocks, while monetary news explains less than 10 percent. The two risk premium components move stocks and bonds differently: the hedging premium pushes bond prices up when stock prices fall, maintaining the hedge; the common premium moves both asset classes in the same direction, destroying it. This asset-price decomposition produces a high-frequency picture of the regime, but it remains backward-looking and cannot explain why risk premia shift. Pflueger (2025) extends CPV to the post-pandemic episode with a critical qualification: supply shocks alone are not sufficient to flip the stock-bond correlation. A hawkish monetary policy stance is also required. Nominal bond-stock betas stayed negative throughout 2021 despite 8 percent CPI inflation, and only turned positive in the second half of 2023 after the Fed completed its tightening cycle. Her model shows that the real-rate channel of supply shocks requires an active monetary response to generate the regime flip. This qualification matters directly for interpreting our sub-period results in Section 5.

The price of inflation risk across asset classes. Beyond the aggregate stock-bond relationship, inflation shapes the entire cross-section of asset returns. But not all inflation is priced equally. Fang, Liu, and Roussanov (2022), hereafter FLR, decompose headline CPI into core inflation which is persistent, wages-and-services-driven and energy inflation which is volatile and

commodity-driven. They show that core inflation betas are consistently negative across all asset classes: stocks, bonds, currencies, REITs, and commodities all fall when unexpected core inflation rises, because core inflation signals lower future real growth and lower dividends. Energy inflation betas are positive on average for real assets, but the price of energy inflation risk is statistically unstable across time. FLR attribute this instability explicitly to variation in whether energy inflation reflects demand or supply, citing Kilian (2009) and Baumeister and Hamilton (2019), but they have no real-time instrument to operationalise this conjecture. They estimate the energy premium using a single structural break in 1999. Section 6 of this paper replaces that break with $\text{energy_}R_t$, a continuous, news-based energy regime indicator constructed from the oil, natural gas, heating oil, and gasoline sub-indices of LMPRP, and tests whether the sensitivity of each asset class to energy inflation shocks varies with the commodity demand regime in a conditional Fama-MacBeth framework across seven asset classes.

3. Data and Construction of the Regime Indicator

Our empirical strategy rests on R_t , a monthly indicator available at the end of each month, before any official macro data is released, that measures whether commodity markets have recently been driven more by demand or by supply forces. Constructing it requires two steps. First, we draw on the LMPRP text-based decomposition to produce monthly NetDemand and NetSupply series. Second, we combine these into a single bounded indicator. We then describe the macro variables that enter the regression controls. Throughout, we take care to distinguish what the data directly measures from what we infer from it, so that a reader unfamiliar with this literature can follow the logic.

3.1 The LMPRP Text-Based Decomposition: How the Indices Are Built

The raw input is a corpus of more than 1,100,000 Reuters news articles published between May 2000 and May 2023, sourced through the Factiva database. After removing near-duplicate articles (two articles sharing more than 70% of identical sentences within a five-day window count as duplicates) the final corpus retains 1,059,669 articles. These articles cover the full commodity spectrum, covering energy, industrial metals, precious metals, agricultural commodities, and livestock. Before any analysis begins, each article is segmented into sentence-level units, and stop

words which are high-frequency function words such as "the", "is", and "at" that carry no economic information are stripped from each sentence.

The construction follows a two-stage procedure. In the first stage, the algorithm identifies supply and demand vocabulary at the sentence level. The core supply dictionary includes roots such as *suppl*, *produc* and *output*; the core demand dictionary includes *demand*, *consum*, *buy* and *purchas*. Individual commodity indices build on this foundation with market-specific terms: crude oil supply adds *rig*, *reserv* and *glut*; wheat supply adds *crop*, *farm* and *harvest*. The algorithm counts co-occurrences where a supply or demand word appears alongside an increase or decrease term within a defined sentence-level proximity window. In the second stage, a refinement layer corrects for semantic inversions: "high temperature" in an agricultural article signals a supply contraction, not an expansion, and "long maintenance" in an energy article means a facility is offline. Commodity-specific exception rules and negation handling ensure that economic meaning, not surface grammar, drives the classification.

The proximity rule that generates the raw counts is straightforward. For each sentence in each article, the algorithm asks: does a supply- or demand-related word appear within $i = 4$ words of a synonym of "increase" or "decrease"? Setting $i = 4$ means that "oil production fell" would count as a supply-decrease signal ("production" is a supply word, "fell" is a decrease synonym, and they are adjacent), while "supply chain disruptions boosted demand for metals" would count as a demand-increase signal. Denoting by N_t the total number of articles published in month t , the four raw counts are :

$$SI_t = N_t^{-1} \sum_w \mathbf{1}[w = \text{Supply}] \cdot \mathbf{1}[|w - \text{posInc}| \leq 4] \quad (1)$$

$$SD_t = N_t^{-1} \sum_w \mathbf{1}[w = \text{Supply}] \cdot \mathbf{1}[|w - \text{posDec}| \leq 4] \quad (2)$$

$$DI_t = N_t^{-1} \sum_w \mathbf{1}[w = \text{Demand}] \cdot \mathbf{1}[|w - \text{posInc}| \leq 4] \quad (3)$$

$$DD_t = N_t^{-1} \sum_w \mathbf{1}[w = \text{Demand}] \cdot \mathbf{1}[|w - \text{posDec}| \leq 4] \quad (4)$$

Dividing by N_t , the total article count, is essential. Without it, the raw counts grow mechanically as Reuters expands its publication volume over time, confounding genuine economic signals with editorial expansion. The denominator converts each count into a *share* of total coverage, making the series comparable across the full 22-year period.

From these four corrected counts, LMPRP compute two net indicators :

$$NetDemand_t = DI_t - DD_t \quad (5)$$

$$NetSupply_t = SD_t - SI_t \quad (6)$$

The sign convention is symmetric and economically consistent. A positive $NetDemand_t$ means demand-increase language outpaces demand-decrease language: reporters are writing more about surging consumption and buying activity than about declining demand. A positive $NetSupply_t$ means supply-decrease language outpaces supply-increase language: reporters are writing more about production cuts and outages than about increased output. Crucially, both a positive $NetDemand$ and a positive $NetSupply$ push commodity prices upward: one through strengthening demand, the other through contracting supply. This is precisely the identification problem that motivates the entire literature: two routes to the same price increase, with opposite macroeconomic implications. The text-based indices resolve this ambiguity by reading the *source* of each movement directly from what journalists write.

The most important property of these indices lies in how they relate across commodities. A principal component analysis of the individual net demand indices reveals a dominant first eigenvalue, a common factor that tracks the global business cycle. Supply shocks tell the opposite story: the scree plot shows no dominant eigenvalue and no natural break. OPEC sets oil quotas, extreme weather destroys wheat harvests, strikes halt copper mining and each event hits one market at a time. This asymmetry, demand shocks common and correlated across markets and supply shocks idiosyncratic and fragmented, determines how commodity disturbances pass through to consumer prices. The indices also pass a battery of validation tests. Human auditors achieve a quarterly correlation of 0.85 with the algorithm's classifications. When major earthquakes strike Chile, the copper supply-decrease index spikes by 1.5 to 1.9 standard deviations the following trading day. Prices react when the news arrives, not before. And when LMPRP compare their oil supply index to the structural benchmarks of Baumeister and Hamilton (2019)

and Känzig (2021), the impulse responses look nearly identical across all three methods, even though the textual approach uses no price or quantity data at all.

The composite index covers all commodities in the S&P Goldman Sachs Commodity Index universe. In our dataset, the composite NetDemand and NetSupply series are standardised to mean zero and standard deviation one, covering 266 monthly observations from May 2001 to June 2023. These are the raw inputs we use to construct R_t . Individual commodity sub-indices (oil, natural gas, heating oil, copper, aluminium, zinc, nickel, wheat, corn, soybeans, sugar, cocoa, coffee, cotton, cattle, and hogs) are also available and enter the energy-specific and composite regime indicators constructed in Section 6.

3.2 Constructing R_t : the Demand Share of Commodity News Volatility

We want a single monthly number that summarizes whether demand or supply has been the dominant force in commodity markets recently. Taking the ratio of levels defined such as $\text{NetDemand}_t / \text{NetSupply}_t$ is poorly behaved because both series pass through zero, making the ratio undefined or explosive in quiet months. A more robust approach is to look at the *relative volatility* of the two signals over a trailing window.

The intuition is direct. In a demand-dominated environment, the demand signal oscillates widely from month to month, with large positive readings when activity accelerates and large negative readings when it cools, while the supply signal is comparatively quiescent. In a supply-dominated environment, the pattern reverses: supply news is volatile (frequent production cuts, outages, geopolitical disruptions) while demand news is more stable. The ratio of rolling standard deviations therefore captures which signal has been the dominant source of variation in commodity news, without being distorted by the direction of any single month's reading. Formally, we define:

$$R_t = \frac{\sigma_{24}(\text{NetDemand}_t)}{\sigma_{24}(\text{NetDemand}_t) + \sigma_{24}(\text{NetSupply}_t)} \quad (7)$$

where σ_{24} denotes the computed over the trailing 24-month window. Using the trailing 24-month standard deviation means that at any date t , R_t reflects the relative volatility of the two news signals over the preceding two years.

R_t is bounded between 0 and 1 by construction, because both σ values are non-negative and the formula is a ratio to their sum. A value close to 1 means demand news has been more volatile than supply news: a demand regime. A value close to 0 means supply news has been more volatile: a supply regime. The natural threshold of 0.5, equal volatility on both sides, divides the two regimes and appears as a dashed horizontal line in Figure 1.

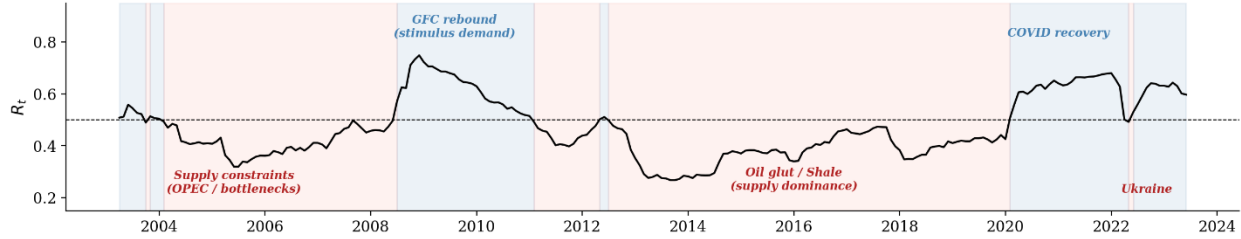
One conceptual precision matters before we proceed. R_t does not capture the direction of demand. It captures whether demand or supply is the dominant source of volatility in commodity news. A high R_t does not automatically mean "inflation is high and output is high." It means demand news dominates commodity price dynamics. Demand can work in two directions simultaneously, and both directions produce the same positive inflation-output covariance. This is the correct reading of the CPV mechanism: what matters is not whether demand news is positive on any given day, but whether demand disturbances are the engine of commodity market variation over the recent past.

The 24-month window requires a minimum of 24 months of data to initialise, so the first valid observation of R_t is April 2003, giving $N = 243$ monthly observations through June 2023. Descriptive statistics computed confirm: mean = 0.470, standard deviation = 0.117, minimum = 0.267, maximum = 0.748. The unconditional correlation between R_t and ρ_t over the full sample is +0.43, already consistent with the CPV prediction before any regression is run.

Figure 1 plots R_t over its full history. Four episodes stand out and each aligns with commodity market events independently documented in the literature. Between 2005 and 2007, R_t drops persistently toward 0.30: OPEC production constraints and Venezuelan output declines generated intense supply-side news coverage even as global demand remained strong (China boom). From mid-2008, R_t rises sharply, reaching its sample maximum of 0.748 in December 2008, three months after Lehman Brothers. This reflects the extreme volatility of demand-side news generated by the GFC demand collapse: as global activity cratered, Reuters coverage of demand conditions became highly variable, which the rolling standard deviation registers as demand-news dominance even in a moment of demand collapse. R_t then remains elevated above 0.64 throughout all of 2009 as fiscal stimulus and Chinese demand recovery sustain intense demand-related coverage. From 2013 onward, the US shale revolution and the OPEC oil glut of 2014–2016 flood supply news, returning, R_t to a prolonged supply regime reaching its sample

minimum in September 2013. The COVID recovery of 2020–2021 lifts the indicator sharply as demand-surge news dominates, before the Ukraine invasion of February 2022 collapses it again. These four transitions emerge from the data without any judgment call on our part.

FIGURE 1. News-Based Demand/Supply Regime Indicator — R_t



Note. The figure plots R_t over the sample period April 2003 – June 2023. R_t is constructed as: $R_t = \sigma_{24}(\text{NetDemand}_t) / [\sigma_{24}(\text{NetDemand}_t) + \sigma_{24}(\text{NetSupply}_t)]$, where σ_{24} denotes the standard deviation computed over the trailing 24-month window, applied to the standardized composite NetDemand and NetSupply series from LMPRP (2025). $R_t \in (0,1)$. Values above 0.5 (blue shading) indicate demand-dominated months; values below 0.5 (red shading) indicate supply-dominated months. The dashed horizontal line marks the 0.5 threshold. Source: own construction from Lumbanraja, Mouabbi, Passari, and Rousset Planat (2025) indices. $N = 243$.

3.3 Constructing ρ_t : the Rolling Inflation-Output Correlation

The dependent variable in Section 4 is a monthly proxy for the CPV macro regime. We define $\rho^{\pi u}_t$ as the 24-month rolling Pearson correlation between CPI inflation (year-on-year) and the unemployment gap. Formally:

$$\rho_t = \text{Corr}_{24}(\pi_t^{\text{YoY}}, \text{ugap}_t) \quad (8)$$

where $\text{ugap}_t = \text{NAIRU}_t - \text{UNRATE}_t$ is a positive-output-gap proxy. The Pearson correlation between two series measures how strongly they co-move in the same direction: +1 means they move perfectly together, −1 means they move perfectly in opposite directions, and 0 means they are unrelated. So, a positive ρ_t means inflation and the unemployment gap move together: a demand-dominated world where the economy runs hot and both inflation and employment are high. A negative ρ_t means inflation rises as the unemployment gap falls: a supply-dominated world, the stagflationary regime where CPV predict the bond hedge breaks.

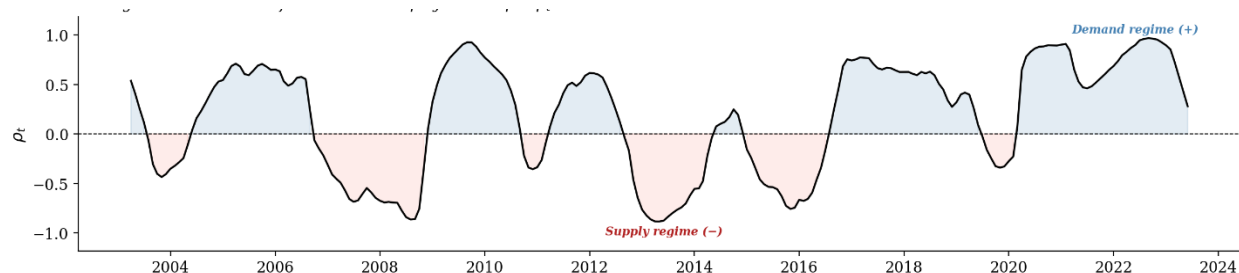
CPI inflation is the year-on-year growth rate of headline CPI (FRED series CPIAUCSL): $(\text{CPI}_t / \text{CPI}_{t-12} - 1) \times 100$. This measures how much prices have risen compared to the same month one year earlier. The **unemployment gap** is defined as $\text{ugap}_t = \text{NAIRU}_t - \text{UNRATE}_t$, where NAIRU (FRED series NROU, Congressional Budget Office estimate) is the non-accelerating

inflation rate of unemployment, meaning the level at which the economy is neither overheating nor in recession, linearly interpolated from its original quarterly frequency to monthly. The sign convention follows CPV: $ugap$ is *positive* when unemployment is below the NAIRU, meaning the economy is running above potential and labor markets are tight; negative when unemployment exceeds NAIRU, meaning slack exists.

We use the unemployment gap rather than a GDP-based output gap for two practical reasons. First, the unemployment rate is published monthly with minimal revision, whereas GDP-based output gap estimates are quarterly and subject to substantial ex-post revisions that compromise real-time analysis. Second, Okun's Law establishes a tight empirical link between unemployment and output at business cycle frequencies, making $ugap$ a reliable proxy. The 24-month rolling correlation between inflation and $ugap$, which we call $\rho_t^{\pi u}$, is positive when both variables rise and fall together (demand-dominated regime: a tight labor market accompanies high inflation) and negative when they move in opposite directions (supply-dominated or stagflationary regime: inflation rises as the economy weakens).

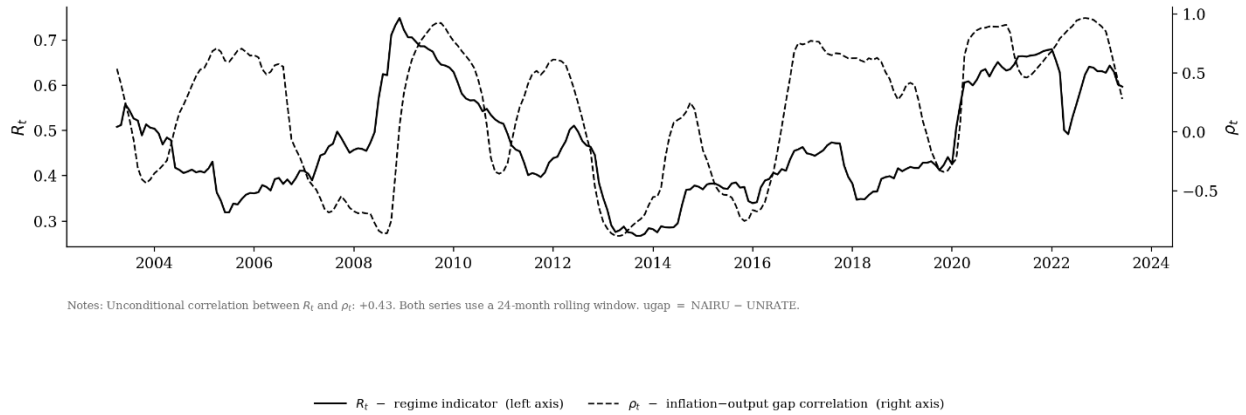
Descriptive statistics for ρ_t : mean = 0.175, standard deviation = 0.564, minimum = -0.883 , maximum = $+0.966$. The wide range confirms the existence of two clearly distinct macro regimes within our sample. The unconditional correlation between R^t and $\rho_t^{\pi u}$ is $+0.43$, computed directly from the data (April 2003 – June 2023, $N = 243$). Figure 2 plots $\rho_t^{\pi u}$; Figure 3 superimposes both series.

FIGURE 2. Rolling Correlation Between CPI Inflation and the Unemployment Gap — ρ_t



Note : The figure plots ρ_t , the 24-month rolling Pearson correlation between year-on-year CPI inflation (FRED: CPIAUCSL) and the unemployment gap ($ugap = NAIRU - UNRATE$, where NAIRU is FRED: NROU linearly interpolated to monthly frequency). The Pearson correlation measures the degree of linear co-movement between two series: $+1$ = move perfectly together; -1 = move perfectly in opposite directions. Positive values (blue shading) correspond to demand-dominated regimes in which inflation and labor market tightness co-move, as in Campbell, Pflueger, and Viceira (2020). Negative values (red shading) indicate supply-dominated or stagflationary regimes. Sample: April 2003 – June 2023. $N = 243$. Descriptive statistics: mean = 0.175, std = 0.564, min = -0.883 , max = $+0.966$. Source: own construction from FRED CPIAUCSL, NROU, UNRATE.

FIGURE 3. Regime Alignment: R_t and ρ_t (Unconditional Correlation = +0.43)



Note : The figure superimposes R_t (solid line, left axis) and ρ_t (dashed line, right axis) over the period April 2003 – June 2023. Both series are constructed using trailing 24-month windows. The unconditional Pearson correlation between the two series is +0.43, computed from $N = 243$ monthly observations. R_t measures the demand share of commodity news volatility (bounded in $(0,1)$); ρ_t measures the rolling inflation-output correlation (bounded in $(-1,+1)$). Source: own construction from LMPRP (2025) indices, FRED CPIAUCSL, NROU, UNRATE.

3.4 Macroeconomic Controls

All regressions include a set of controls capturing the broader macro environment. Monetary policy rate: the Wu-Xia (2016) shadow rate, spliced with the federal funds rate (FRED: FEDFUNDS). Because the federal funds rate hit its zero lower bound from December 2008 onward and conveys no information about marginal policy tightness, we replace it with the Wu-Xia shadow rate from July 2009 through December 2015, which remains negative during QE and correctly captures the stimulative effect of unconventional policy, then revert to the actual rate thereafter. Dollar index: trade-weighted broad dollar index, spliced from FRED's TWEXB (weekly, available through 2005) and DTWEXBGS (daily, from 2006), both aggregated to monthly means and merged on their overlapping window. VIX: CBOE Volatility Index (FRED: VIXCLS), aggregated to monthly means from daily observations, controlling for equity market stress. Term spread: 10-year minus 2-year Treasury constant-maturity yield (FRED: T10Y2Y), aggregated to monthly means from 6,782 daily observations (missing values on weekends and holidays excluded before aggregation). The final merged dataset covers January 1999 to December 2024, $N = 312$ monthly observations.

3.5 The stock-bond correlation ρ_t^{SB}

We construct ρ_t^{SB} as the 24-month rolling Pearson correlation between monthly S&P 500 total returns and monthly approximate 10-year Treasury returns. Formally:

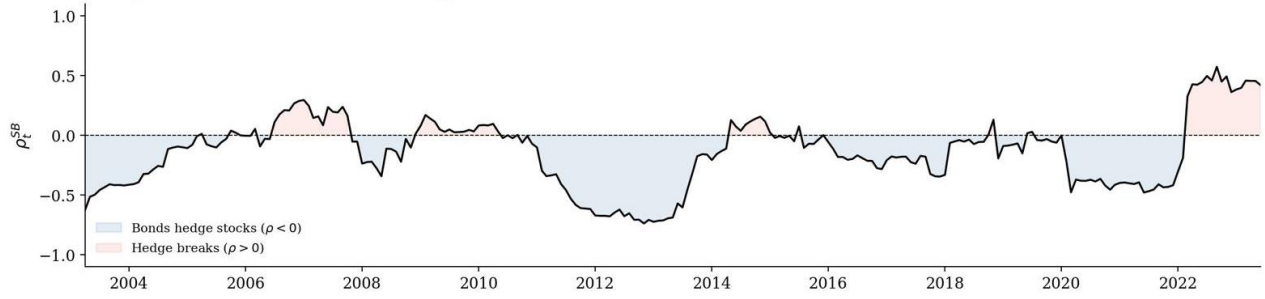
$$\rho_t^{SB} = \text{Corr}_{24}(r_t^{\text{SP500}}, r_t^{\text{bond}}) \quad (9)$$

Stock returns are computed from the S&P 500 Total Return Index (Bloomberg SPXT, extracted from): $\text{ret}_t = (\text{SPXT}_t / \text{SPXT}_{t-1}) - 1$. Bond returns are approximated from FRED's DGS10 (daily 10-year constant-maturity yield, 6,782 daily observations with missing values on weekends excluded before monthly aggregation) using the modified-duration formula: $\text{bond_ret}_t = y_{t-1}^{10}/12 - 9.0 \times \Delta y_t^{10}$. The coefficient 9.0 approximates the modified duration of a 10-year par bond.

The resulting series runs from April 2003 to June 2023, giving $N = 243$ monthly observations, co-incident with the valid sample for R_t . Over the full sample, ρ_t^{SB} has a mean of -0.142 and a standard deviation of 0.287 , confirming that the post-2001 period is predominantly one where bonds hedged equities. The series ranges from -0.739 in November 2012, the height of the post-GFC quantitative easing regime, to $+0.574$ in September 2022, at the peak of the Federal Reserve's most aggressive tightening cycle in four decades.

The 2022 episode is particularly striking and central to our thesis. ρ_t^{SB} turns positive in March 2022, the month Russia invaded Ukraine, and stays positive for the remainder of our sample. By September 2022, it reaches its all-time high of $+0.574$. Over the full year 2022, the correlation averages $+0.427$, compared to -0.158 over the 2001 to 2019 period. Stocks and bonds fell together throughout this episode — the 60/40 portfolio delivered its worst calendar-year performance in decades — exactly as the CPV framework predicts for a supply-shock regime. Figure 4 plots ρ_t^{SB} over its full history. Blue shading marks months where bonds hedge equities ($\rho_t^{SB} < 0$); red shading marks months where the hedge breaks ($\rho_t^{SB} > 0$).

FIGURE 4 . 24-Month Rolling Stock-Bond Correlation — ρ_t^{SB}



Note. The figure plots ρ^{SB} , the 24-month rolling Pearson correlation between monthly S&P 500 total returns (Bloomberg SPXT) and monthly approximate 10-year Treasury returns. Bond returns are approximated as $\text{bond_ret}_t = y_{t-1}^{10}/12 - 9.0 \times \Delta y_t^{10}$, where 9.0 is the approximate modified duration of a 10-year par bond and y^{10} is from FRED DGS10. Blue shading: $\rho^{SB} < 0$ (bonds hedge stocks). Red shading: $\rho^{SB} > 0$ (hedge breaks). Sample: April 2003 – June 2023. $N = 243$. Descriptive statistics: mean = -0.142 , std = 0.287 , min = -0.739 (November 2012), max = $+0.574$ (September 2022). Source: own construction.

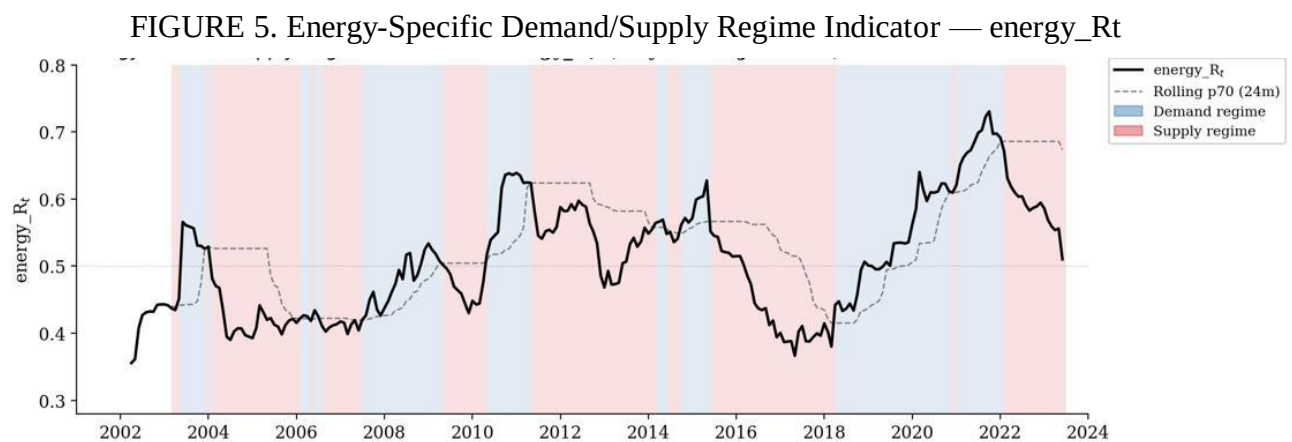
3.6. Asset returns and Inflation Shocks for Cross-Sectional Tests

We build energy_R_t from the four energy commodity sub-indices of the LMPRP data: oil, natural gas, heating oil, and gasoline. For each commodity, the data provides monthly standardized NetDemand and NetSupply signals constructed by the same proximity-rule algorithm described in Section 3.1. We average these four commodities into a single monthly energy demand signal $\text{NetDemand}^{\text{energy}}_t$ and $\text{NetSupply}^{\text{energy}}_t$, then apply the same rolling-volatility ratio formula used to build the composite R_t in Section 3.2.

The indicator is bounded in $(0, 1)$: values near 1 signal demand dominance in energy markets, values near 0 signal supply dominance. Over the analysis window May 2001 – June 2023, energy_R_t has mean 0.511, standard deviation 0.084, minimum 0.366, and maximum 0.731 (October 2021, peak of the post-COVID demand surge). The correlation with the composite R_t is 0.42, confirming that the two indicators capture related but distinct signals: energy_R_t focuses specifically on the four energy commodities that enter CPI energy inflation, while R_t covers the full commodity spectrum. Four historical episodes align correctly with known commodity market events: the shale glut of 2014–2016 is classified as a supply regime (31% demand months), the 2020–2021 COVID recovery as a demand regime (95% demand months), the 2022 Ukraine invasion as a supply regime (8% demand months), and the 2008–2009 GFC as a demand regime driven by extreme volatility in demand-side coverage (89% demand months). These classifications

match the episodes as documented by Kilian (2009) and Känzig (2021) for oil, and by LMPRP (2025) for the broader commodity spectrum.

Figure 5 plots energy_R_t over its full history alongside the rolling 24-month 70th-percentile threshold used to classify demand-dominated months. The indicator captures the deep supply cycle of 2014–2016 (OPEC oil glut and US shale production growth), the rapid demand rebound of 2020–2021, and the abrupt shift to supply dominance following the February 2022 Ukraine invasion. The demand dummy, which equals 1 when energy_R_t exceeds this rolling 70th percentile, classifies 103 months as demand-dominated (42%) and 141 as supply-dominated (58%) over the March 2003 – June 2023 analysis window ($N = 244$).



Note. The figure plots energy_R_t — the 24-month rolling demand-share of volatility in energy commodity news (oil, natural gas, heating oil, gasoline) — over May 2001 – June 2023, constructed from the LMPRP (2025) sub-indices. Blue background: demand regime (energy_R_t above rolling 24-month 70th percentile). Red background: supply regime. The dashed line marks the rolling 70th-percentile threshold. $N = 244$ (March 2003 – June 2023). Key episodes: shale glut 2014–2016 (supply), COVID recovery 2020–2021 (demand), Ukraine 2022 (supply). Source: own construction.

Section 6 tests whether the commodity demand regime explains the cross-sectional pricing of inflation risk. This requires two inputs: a broad universe of asset class returns, and a time series of structural inflation shocks.

Asset returns. We build a panel of 35 monthly return series spanning seven asset classes, following the portfolio construction of Fang, Liu, and Roussanov (2022). All series are excess returns in percent per year and enter our dataset, which covers May 2001 to June 2023. Stocks (s1 to s5) are five US industry portfolios from the Kenneth French data library: Consumer (s1),

Manufacturing (s2), High Tech (s3), Health (s4), and Other (s5) and we construct them as value-weighted excess returns over the risk-free rate. Treasuries (dgs1 to dgs30): eight approximated total return series for the 1, 2, 3, 5, 7, 10, 20, and 30-year constant-maturity yields from FRED, using the Gürkaynak, Sack, and Wright (2007) duration-based approximation. Corporate bonds (cp1 to cp4): four Bloomberg US Investment Grade corporate bond total return indices spanning the maturity spectrum. REITs (re1 to re3): three FTSE NAREIT total return indices: Equity REITs (re1, FNERTR), Mortgage REITs (re2, FNMRTTR), and All REITs (re3, FNAR). International stocks (in1 to in3) are three MSCI total return indices in USD: North America (in1, NDDUNA), Europe (in2, NDDUEURO), and Far East (in3, NDDUFE). Commodities (cm1 to cm5): five S&P GSCI equal-weighted sector total return indices. Currencies (fx1 to fx7): seven carry-sorted currency portfolios from Lustig, Roussanov, and Verdelhan (2011), available from January 1999 to May 2021 ($N = 241$). fx1 is the equal-weighted dollar carry factor (RX) and fx2 through fx7 are carry portfolios sorted from lowest to highest interest rate.

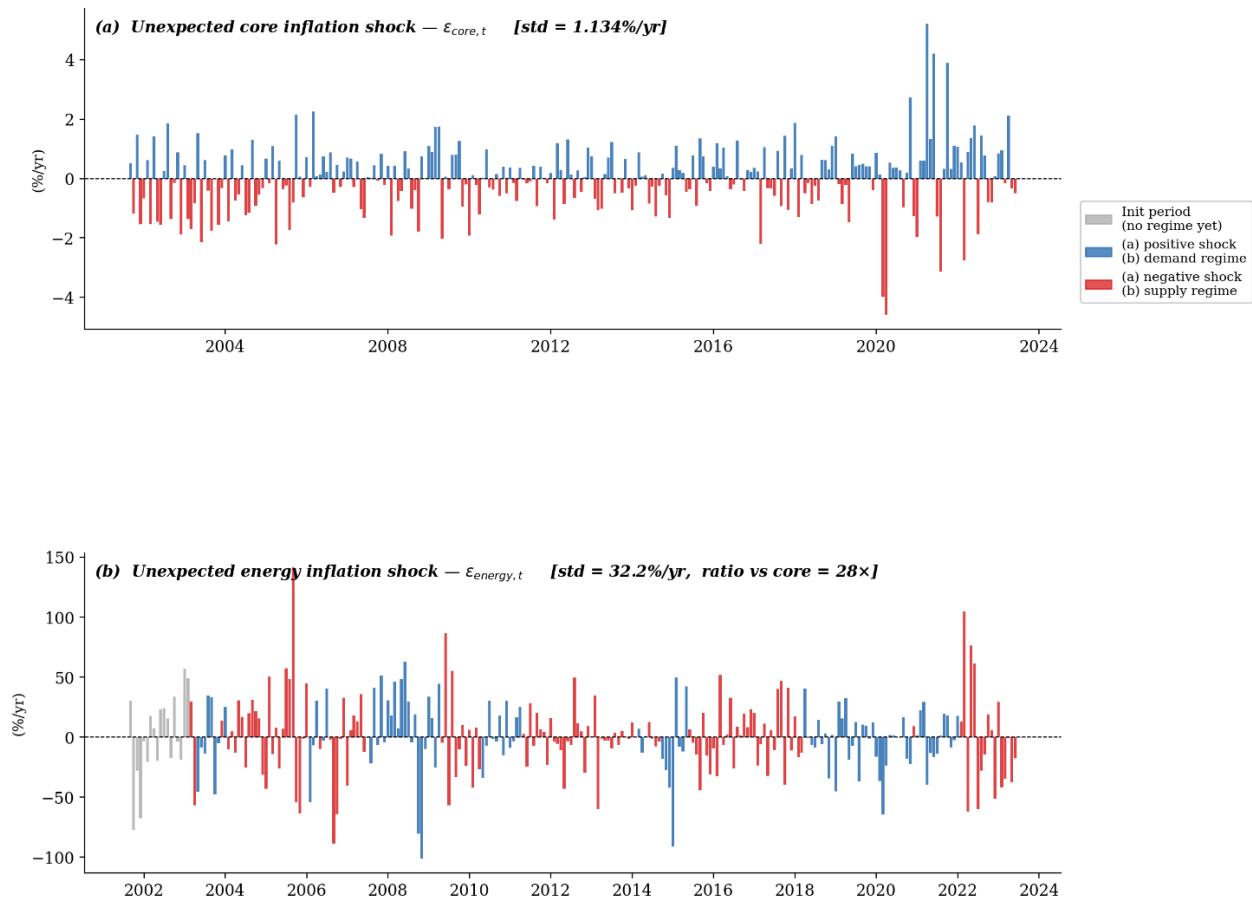
Asset prices react to *unexpected* changes in macroeconomic variables. An inflation reading that exactly matches what the market anticipated carries no new information and should produce no asset price movement. To isolate the genuinely unexpected component of inflation, we follow FLR (2022) and estimate a six-variable VAR(4). The system contains, in order: monthly annualized core CPI inflation π^{core}_t , food CPI inflation π^{food}_t , energy CPI inflation π^{energy}_t , the risk-free rate (Wu-Xia shadow rate spliced with the federal funds rate), the log price-dividend ratio (Shiller, 2024), and the unemployment gap. All three inflation series are computed as month-over-month annualized percentage changes, $\pi_t = (CPI_t / CPI_{t-1} - 1) \times 12 \times 100$, using FRED series CPILFESL, CPIFABSL, and CPIENGSL respectively. We use month-over-month rather than year-over-year inflation because year-over-year changes are a 12-month moving average that smooths away genuine monthly surprises. The VAR is estimated over January 1999 – December 2024, using the full available history to maximize the precision of the forecast, even though our asset pricing analysis is restricted to the March 2003 – June 2023 window.

The reduced-form VAR residuals are the monthly inflation surprises. Over the March 2003 – June 2023 analysis window, ε^{core}_t has mean 0.02 and standard deviation 1.13 %/year, while ε^{energy}_t has mean approximately zero and standard deviation 32.0 %/year, making energy inflation shocks about 28 times more volatile than core inflation shocks, consistent with the well-documented

commodity price volatility in Kilian (2009). Following FLR, we use the raw reduced-form residuals and rely on the cross-sectional spread of betas across asset classes for identification.

Figure 6 plots both shock series over the analysis window. Panel (a) shows $\varepsilon_{core,t}$ in blue (positive shocks) and red (negative shocks). Panel (b) shows $\varepsilon_{energy,t}$ with bars colored by demand regime (blue) and supply regime (red), revealing the regime structure that the conditional Fama-MacBeth test exploits. The gray bars in panel (b) correspond to months before the rolling threshold produces a valid regime classification (May 2001 – February 2003).

FIGURE 6. VAR(4) Inflation Shocks: Unexpected Core and Energy CPI Inflation



Note. Panel (a): unexpected core inflation shock $\varepsilon_{core,t}$ (residual from the core CPI equation of the VAR). Blue: positive shock; Red: negative shock. Panel (b): unexpected energy inflation shock $\varepsilon_{energy,t}$. Blue: demand-regime month (energy_Rt above rolling 24-month 70th percentile); Red: supply-regime month; Gray: initialization period (regime dummy not yet available). Both shocks are VAR(4) reduced-form residuals following Fang, Liu, and Roussanov (2022). MoM annualized inflation $\pi_t = (CPI_t / CPI_{t-1} - 1) \times 12 \times 100$. $\sigma(\varepsilon_{core}) = 1.13$ %/yr; $\sigma(\varepsilon_{energy}) = 32.0$ %/yr. Sample: May 2001 – June 2023. Source: own estimation from FRED CPILFESL, CPIFABSL, CPIENGSL.

4. Does Commodity News Predict the Macroeconomic Regime?

4.1 Regression Framework: Local Projections

To test whether R_t has genuine predictive content for the inflation-output regime, we estimate local projections following Jordà (2005). A local projection is a direct regression of the future value of the outcome variable on current predictors, estimated separately for each forecast horizon h . For each $h \in \{3, 6, 12\}$ months:

$$\rho_{t+h}^{\pi u} = \alpha_h + \beta_h \cdot R_t + \gamma_h' X_t + \varepsilon_{t+h} \quad (10)$$

where X_t includes current CPI inflation, the spliced policy rate, and the lagged dependent variable $\rho^{\pi u}$. The lagged dependent variable is included because the 24-month rolling correlation is highly autocorrelated by construction, so any predictor must add information beyond the autoregressive benchmark to claim genuine predictive content.

Because $\rho_{t+h}^{\pi u}$ uses data from t to $t+h$ and thereby overlaps with $\rho_{t+h-1}^{\pi u}$, the residuals ε_{t+h} are serially correlated by construction with a moving-average structure of order h . Standard OLS standard errors would be too small. We correct using Newey-West HAC standard errors with bandwidth $h+1$, a heteroskedasticity- and autocorrelation-consistent correction that remains valid even when the exact error structure is unknown. The advantage of local projections over iterated VARs is that each horizon is estimated directly, without compounding model misspecification across horizons. Plagborg-Møller and Wolf (2021) show that local projections and VARs estimate the same impulse responses in population, so there is no loss of information from this approach.

4.2 Full-Sample Results

Table 1 reports the full-sample estimates. over 243 monthly observations from April 2003 to June 2023. The base sample has 243 monthly observations from April 2003 to June 2023, but the local projection regressions use slightly smaller samples at longer horizons because projecting h months ahead consumes the last h observations: $N = 240$ at $h = 3$, $N = 237$ at $h = 6$, and $N = 231$ at $h = 12$. The coefficient on R_t is positive at all three horizons and grows monotonically with h : $\hat{\beta} = 0.39$ ($p = 0.27$) at $h = 3$, $\hat{\beta} = 0.94$ ($p = 0.09$) at $h = 6$, and $\hat{\beta} = 2.02$ ($t = 3.61$, $p < 0.001$, 95% CI: $[0.92, 3.12]$) at $h = 12$. The monotonically increasing profile is the key result: R_t does not merely move contemporaneously with the macro regime. It leads it. A one-standard-deviation increase in R_t (0.117 units) predicts $\rho_{t+12}^{\pi u}$ to be 0.24 higher twelve months later, a shift

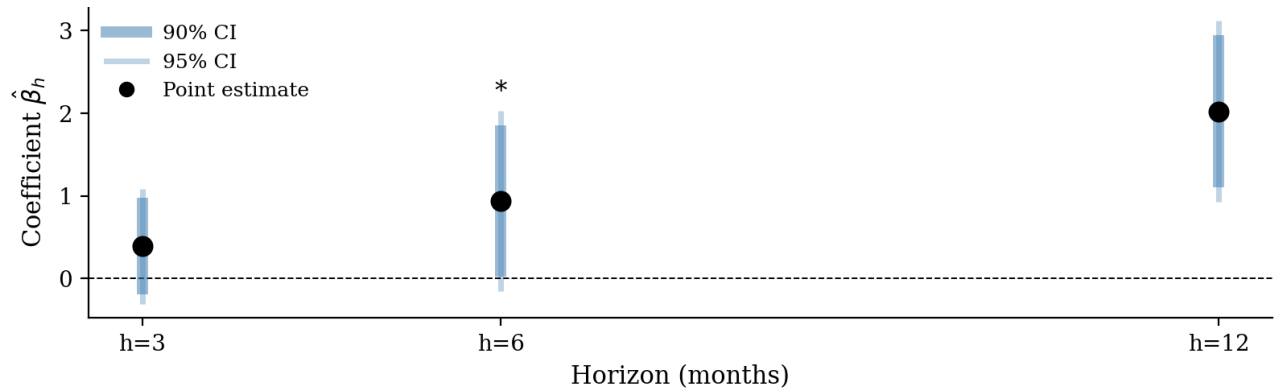
of approximately 0.42 standard deviations of $\rho_t^{\pi u}$, which is economically large. The R^2 values in Table 1 deserve a word of interpretation. At $h = 3$, $R^2 = 0.77$, which is high, but this reflects the strong autocorrelation of the rolling correlation itself rather than the predictive power of R_t alone as the lagged ρ_t dominates at short horizons. As the horizon extends to $h = 12$, R^2 falls to 0.17, and the contribution of R_t relative to the autoregressive benchmark grows. The monotonic rise in $\hat{\beta}_h$ alongside the falling R^2 confirms that R_t adds genuinely forward-looking information that the lagged dependent variable can not provide.

TABLE 1. Local Projection Results — Full Sample, Dependent Variable: ρ_{t+h}

Horizon	Coeff. R_t	t -stat	p -value	R^2	N
$h = 3$	0.392	(1.10)	0.271	0.766	240
$h = 6$	0.939*	(1.68)	0.092	0.431	237
$h = 12$	2.022***	(3.61)	<0.001	0.172	231

Note. The table reports local projection estimates of ρ_{t+h} on R_t , where ρ_t is the 24-month rolling Pearson correlation between year-on-year CPI inflation and the unemployment gap, and R_t is the news-based demand share of commodity news volatility (see Section 3). Controls: lagged ρ_t , year-on-year CPI inflation, Wu-Xia shadow rate (spliced with the federal funds rate for July 2009 – December 2015). Newey-West HAC standard errors, bandwidth $h+1$. Sample: April 2003 – June 2023. $N = 240$ ($h=3$), 237 ($h=6$), 231 ($h=12$). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Source: own estimation from our calculations.

FIGURE 7. Impulse Response Function: Effect of R_t on ρ_{t+h} at $h = 3, 6, 12$ Months



Note. The figure plots $\hat{\beta}^h$, the local projection coefficient on R_t in the regression of ρ_{t+h} on R_t and controls, at horizons $h = 3, 6, 12$ months, with 90% (darker) and 95% (lighter) Newey-West confidence intervals (bandwidth $h+1$). A monotonically increasing profile indicates that R_t leads the macro regime rather than being merely contemporaneous with it. Full sample, April 2003 – June 2023. Source: own construction.

4.3 Sub-Sample Analysis: Monetary Policy Regimes Matter

Table 2 and Figure 8 split the sample into three periods following Pflueger (2025): the pre-GFC Greenspan era (April 2003 – December 2008, $N = 57$ at $h=12$), the zero lower bound period (January 2009 – December 2015, $N = 72$ at $h=12$), and the post-ZLB normalization (January 2016 – June 2023, $N = 78$ at $h=12$).

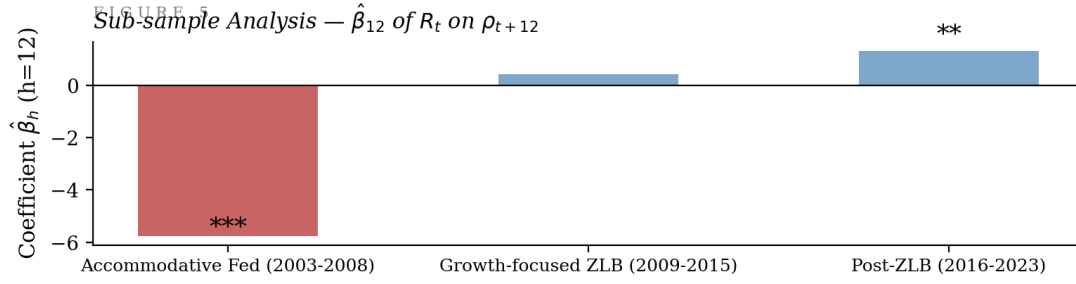
In the Greenspan era (2003 to 2008), $\hat{\beta}_{h=12} = -5.75 (t = -7.14, p < 0.001)$, negative and strongly significant. Far from being a contradiction, this reflects the Fed's pre-emptive tightening strategy: as soon as demand signals in commodity markets picked up and R_t rose, the Fed tightened, suppressing inflation and decoupling it from the unemployment gap. The policy response dominated the commodity channel, so R_t predicted a negative $\rho_t^{\pi u}$ rather than a positive one. This result is entirely consistent with Pflueger (2025), who shows that supply shocks (and by symmetry, demand signals) only flip the macro regime when monetary policy is active enough to transmit them. Greenspan's pre-emptive stance was so aggressive that it broke the link between commodity demand news and the realized inflation-output correlation. During the ZLB period (2009 to 2015), $\hat{\beta}_{h=12} = +0.42 (p = 0.79)$, statistically indistinguishable from zero, consistent with reduced monetary transmission during unconventional policy. In the post-ZLB period (2016 to 2023), $\hat{\beta}_{h=12} = +1.32 (t = 2.29, p = 0.022)$, positive and significant, exactly as CPV predict when monetary policy has room to respond.

TABLE 2. Local Projection Results by Monetary Policy Sub-Period — $h = 12$ Months

Horizon	Accommodative Fed (2003-2008)	Growth-focused ZLB (2009-2015)	Post-ZLB (2016-2023)
$h = 3$	-1.813 (-1.02)	-0.242 (-0.19)	-0.051 (-0.13)
$h = 6$	-5.916*** (-3.51)	0.323 (0.15)	-0.111 (-0.20)
$h = 12$	-5.754*** (-7.14)	0.423 (0.27)	1.317** (2.29)

Note. The table reports $\hat{\beta}$ at $h = 3, 6, 12$ months estimated separately within each monetary policy sub-period, following the taxonomy of Pflueger (2025). Sub-periods: Accommodative/Pre-emptive Fed (April 2003 – December 2008); Zero Lower Bound (January 2009 – December 2015); Post-ZLB normalization (January 2016 – June 2023). Dependent variable: ρ_{t+h} . Controls and standard errors as in Table 1. Observation counts at $h=12$: 57, 72, 78 respectively. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Source: own estimation.

FIGURE 8. β_h Across Monetary Policy Sub-Periods ($h = 3, 6, 12$)



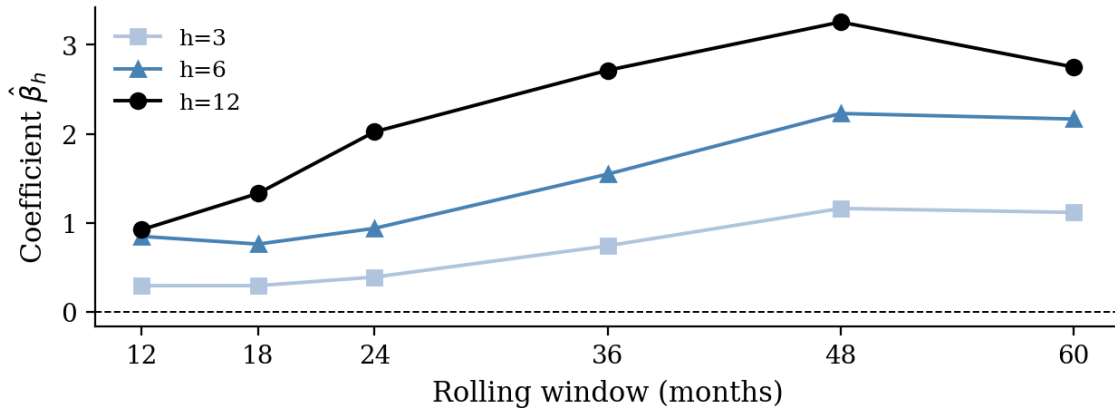
Notes: Blue = positive; Red = negative. Newey-West HAC s.e. * $p < 0.10$ ** $p < 0.05$ *** $p < 0.01$.
 2003-2008 (Accommodative Fed): negative beta consistent with Pflueger (2025) — an accommodative central bank offsets commodity regime dynamics.
 2009-2015 (ZLB): near-zero beta validates the framework — ZLB prevents monetary transmission regardless of the commodity regime.
 2016-2023 (Post-ZLB): positive and significant beta confirms H1 under a normally functioning monetary policy regime.

Note. The figure plots β^h at $h = 3, 6, 12$ months within each monetary policy sub-period. Sub-period taxonomy follows Pflueger (2025). 90% Newey-West confidence intervals. Source: own construction from

4.4 Robustness: Rolling Window Length

Figure 9 repeats the estimation for rolling windows of 12, 18, 24, 36, 48, and 60 months used to construct both R_t and $\rho_t^{\pi u}$. The coefficient at $h = 12$ is positive and significant for windows of 18 months and above ($p = 0.042$ at 18 months; $\hat{\beta} = 2.71$, $p < 0.001$ at 36 months; $\hat{\beta} = 3.26$, $p < 0.001$ at 48 months; $\hat{\beta} = 2.75$, $p = 0.005$ at 60 months). The 12-month window falls below significance ($\hat{\beta} = 0.92$, $p = 0.164$), as expected given the noisiness of very short rolling windows. The fact that the coefficient grows with the window length up to 48 months and then stabilizes suggests that longer windows reduce noise in both R_t and ρ_t , sharpening the signal. The 24-month baseline is a conservative choice. The positive relationship is not an artefact of the 24-month baseline.

FIGURE 9. Robustness: β_h Across Rolling Window Lengths ($h = 3, 6, 12$)



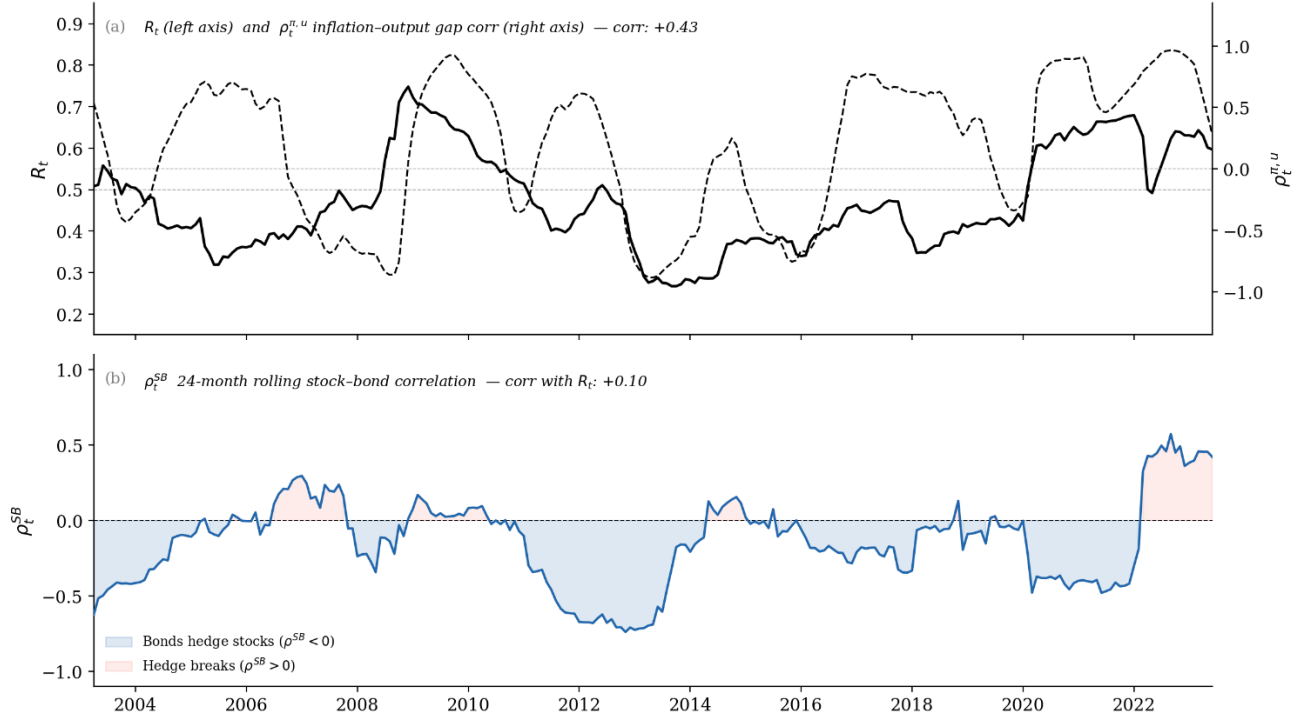
Note. The figure plots β at $h = 3, 6, 12$ months for rolling windows of 12, 18, 24, 36, 48, and 60 months used to construct both R_t and ρ_t . The 24-month baseline is highlighted. Newey-West HAC standard errors (bandwidth $h+1$). Source: own construction.

5. Does the Regime Signal Predict the Stock-Bond Correlation?

5.1 Setup and the Dependent Variable

Section 4 establishes that R_t leads the inflation-output regime. The CPV mechanism then predicts that R_t should also predict the stock-bond correlation ρ_t^{SB} , the direct portfolio consequence. When R_t is high and demand dominates, the inflation-output correlation is positive, the central bank can cut rates when equities fall, and bonds hedge equities: $\rho_t^{SB} < 0$. When R_t is low and supply dominates, the inflation-output correlation turns negative, bonds can no longer rally to offset equity losses, and ρ_t^{SB} turns positive. The test in this section is therefore whether a real-time commodity news signal can deliver the portfolio signal that CPV's theoretical regime variable delivers ex post. Figure 10 provides the visual motivation for this section. Panel (a) plots R_t alongside $\rho_t^{\pi u}$, the rolling inflation-output gap correlation that Section 4 shows R_t leads at a 12-month horizon. The two series co-move positively (unconditional correlation = +0.43): when R_t rises toward demand dominance, the inflation-output correlation tends to follow. Panel (b) shows ρ_t^{SB} , the rolling stock-bond correlation. For most of the sample the correlation is negative, meaning bonds hedge equities.

FIGURE 10. Regime Alignment



Notes. All series use a 24-month rolling window. Panel (a): R_t (left axis, solid black) is the demand share of commodity news volatility from equation (7); $\rho_t^{\pi,u}$ (right axis, dashed) is the rolling Pearson correlation between year-on-year CPI inflation and the unemployment gap. Unconditional correlation between the two series: +0.43. Panel (b): ρ_t^{SB} is the rolling Pearson correlation between monthly S\&P 500 Total Return and the approximated 10-year Treasury return, computed as $r_t^{\text{bond}} = (y_{t-1}/12) - 9 \times \Delta y_t$. Blue shading marks months where bonds hedge stocks ($\rho_t^{SB} < 0$); red shading marks months where the hedge breaks ($\rho_t^{SB} > 0$). Unconditional correlation between R_t and ρ_t^{SB} : +0.10. Sample: April 2003 to June 2023.

5.2 Baseline Regression

We estimate the contemporaneous OLS regression:

$$\rho_t^{SB} = \alpha + \beta \cdot R_t + \gamma' X_t + \varepsilon_t \quad (11)$$

where X_t includes CPI inflation, realized equity volatility computed as the 24-month rolling standard deviation of S\&P 500 returns, the log-change in the dollar index, VIX, the Wu-Xia policy rate, and the term spread. We use Newey-West HAC standard errors with bandwidth 24 throughout. Table 3 reports two specifications.

In Model (1) with R_t alone: $\hat{\beta} = 0.24$, $p = 0.61$, $R^2 = 0.010$. Adding controls in Model (2): $\hat{\beta} = 0.43$, $p = 0.55$, $R^2 = 0.271$. Neither specification produces a significant coefficient. The R^2 improvement in Model (2) comes from the joint explanatory power of the controls, not from R_t itself. The full-sample correlation between R_t and ρ_t^{SB} is $+0.098$, in the wrong direction relative to the CPV prediction that higher R_t should predict more negative ρ_t^{SB} . Section 5.3 explains why this happens and why it is a feature of the 2022–2023 episode rather than a failure of the underlying mechanism.

TABLE 3. Baseline Regressions of ρ_t^{SB} on R_t

Variable	(1) R_t only	(2) R_t + controls
R_t	0.2395 (+0.506)	0.4333 (+0.605)
Inflation (YoY)	—	-0.0002 (-0.006)
Equity vol (24m)	—	-0.8245 (-0.422)
Δ Dollar index	—	1.4462 (+1.473)
VIX	—	0.0014 (+0.252)
Policy rate	—	0.0426 (+1.036)
Term spread	—	-0.0515 (-0.656)
Constant	-0.2546 (-1.113)	-0.2361 (-1.046)
R^2	0.010	0.271
Observations	243	243

Note. The table reports OLS estimates of ρ^{SB} on R_t . Model (1): R_t alone. Model (2): R_t with controls (realized equity volatility, log-change in dollar index, VIX, Wu-Xia policy rate, term spread, CPI inflation). Newey-West HAC standard errors, bandwidth 24. Sample: April 2003 – June 2023. $N = 243$. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Source: own estimation.

5.3 The 2022–2023 Episode and the Structural Constraint

In 15 out of the 16 months from March 2022 to June 2023, R_t was above 0.5 (averaging 0.605, a demand-dominance signal) while ρ_t^{SB} was simultaneously positive (averaging $+0.44$, the worst hedging environment in the sample). In these months, the backward-looking indicator still read demand dominance while the actual regime had already flipped to supply, and this

mechanically flipped the full-sample correlation from -0.07 (excluding all months where both $R_t^i > 0.55$ and $\rho_t^{SB} > 0$) to $+0.098$ (full sample). Excluding the 2022–2023 episode entirely leaves the full-sample correlation negative at -0.069 .

The mechanism is structural. The 24-month backward-looking window incorporated the entire 2020–2021 COVID demand boom, so R_t^i remained elevated in 2022 even as the Ukraine invasion brought a supply shock of the first order. Simultaneously, Ukraine war news generated demand-side coverage (strategic reserve releases, LNG import demand, consumption substitution) alongside supply disruption news, further confounding the composite indicator. Testing rolling windows of 12, 18, 36, and 48 months all produce a positive full-sample correlation. The problem is not the window length but the nature of the episode.

This result is theoretically coherent with Pflueger (2025), who shows that the stock-bond correlation flips only when supply shocks are simultaneously inflationary and confronted by a hawkish monetary policy response. R_t captures the commodity news side of this condition but not the monetary policy response side. The supply-dominated months in the 2003–2019 portion of our sample (the 2014–2016 oil glut, the 2013–2014 commodity price weakness) were deflationary supply shocks. In those episodes the central bank cut rates freely and bonds continued to hedge stocks, so ρ_t^{SB} stayed negative even as R_t signalled supply dominance. A true stagflationary supply shock, one that forces the Fed into a dilemma, only clearly materialized in 2022–2023, and that episode is precisely where the backward-looking rolling window mislabels the regime. We frame this as a finding rather than a failure: it identifies the condition under which a commodity news indicator alone is insufficient and must be combined with a monetary policy signal.

5.4 Predictive Local Projections

Although the contemporaneous regression does not deliver a significant coefficient, R_t may carry forward-looking information that only materializes with a lag. We estimate local projections of ρ_{t+h}^{SB} on R_t at horizons $h \in \{3, 6, 12\}$ months, following the same specification as equation (10) with ρ_t^{SB} as the dependent variable and Newey-West HAC standard errors with bandwidth $\max(24, h)$. Table 4 and Figure 11 report the results. The coefficient on R_t grows monotonically with the horizon: $\hat{\beta} = 0.41(p = 0.48)$ at $h = 3$, $\hat{\beta} = 0.70(p = 0.16)$ at $h = 6$, and $\hat{\beta} = 1.38(t = 1.82, p = 0.068)$ at $h = 12$. **The 90% confidence interval at $h = 12$ lies entirely above zero:**

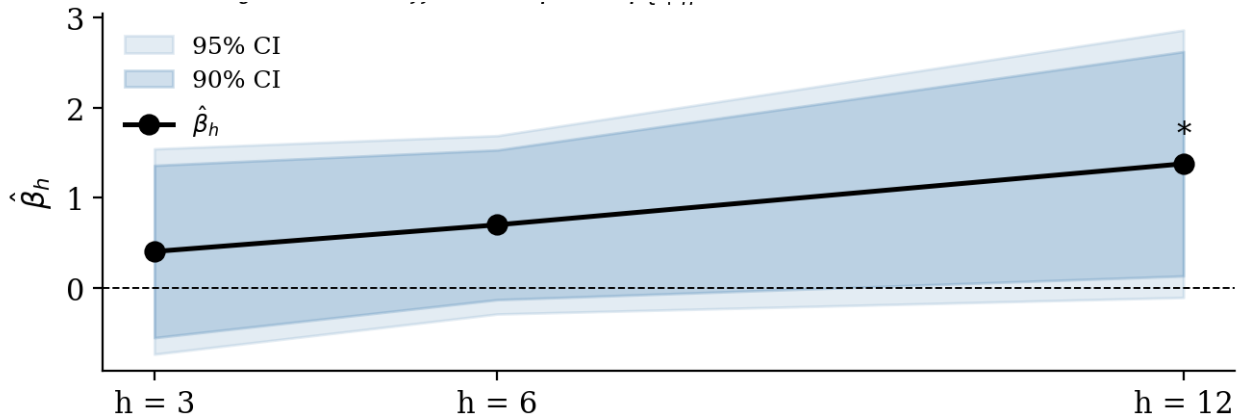
[0.13, 2.62]. The pattern mirrors exactly what we found for the inflation-output regime in Section 4: R_t adds predictive content that grows with the horizon, consistent with a leading indicator rather than a contemporaneous one.

TABLE 4. Local Projections of ρ_{t+h}^{SB} on R_t

Horizon	$\hat{\beta}_h$	t-stat	p-value	R^2	N
h = 3	0.4058	(+0.699)	0.4845	0.243	240
h = 6	0.7014	(+1.391)	0.1642	0.260	237
h = 12	1.3778*	(+1.823)	0.0683	0.316	231
95% CI at h=12: [-0.1036, 2.8592]					

Note. The table reports local projection estimates of ρ_{t+h}^{SB} on R_t and controls at h = 3, 6, 12 months. Controls: lagged ρ^{SB} , CPI inflation, Wu-Xia policy rate, VIX, term spread, dollar change. Newey-West HAC standard errors, bandwidth max(24, h). N = 240, 237, 231 at h = 3, 6, 12. *p < 0.10, **p < 0.05. Source: own estimation.

FIGURE 11. Impulse Response Function: Effect of R_t on ρ_{t+h}^{SB} at h = 3, 6, 12 Months



Note. The figure plots $\hat{\beta}_h$ from local projections of ρ_{t+h}^{SB} on R_t at h = 3, 6, 12 months, with 90% and 95% Newey-West confidence intervals. The monotonically increasing profile confirms that R_t leads the stock-bond correlation rather than tracking it contemporaneously. Source: own construction

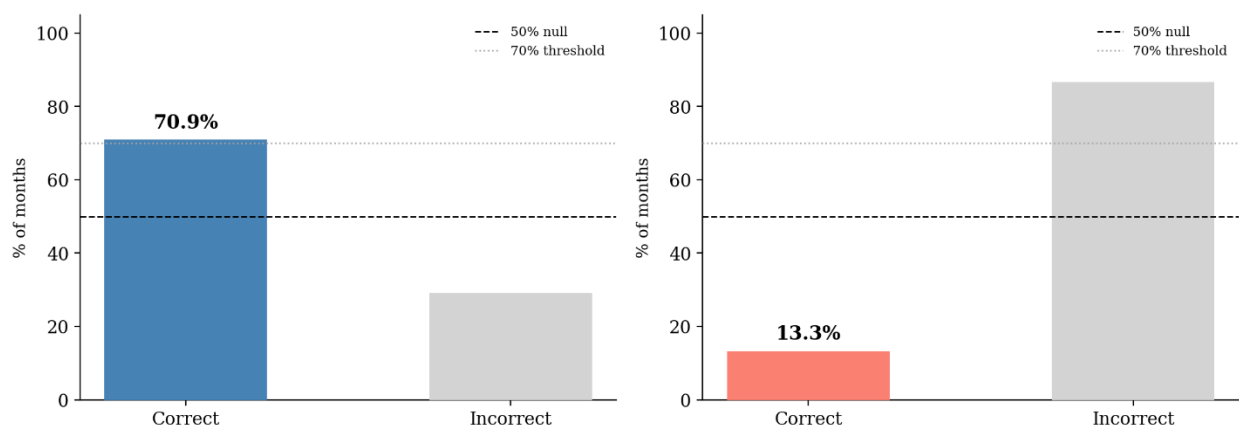
5.5 Regime Classification Analysis

To assess whether R_t has directional accuracy in classifying months as hedging or non-hedging, we define demand-dominated months as those where R_t exceeds its rolling 70th percentile computed over a 60-month backward window, and supply-dominated months as those where R_t falls below the rolling 30th percentile (same backward window). We test each against a

50% coin-flip null using a one-sided binomial test, a test of whether the observed proportion of correct classifications could plausibly arise by chance if R_t contained no information about ρ^{SB}_t . The 60-month rolling window for the percentile thresholds ensures that what counts as "high" or "low" is determined by the local distribution of R_t , not its full-sample distribution, maintaining real-time validity.

Results in demand-dominated months ($N = 86$), $\rho^{SB}_t < 0$ in 70.9% of cases, significantly above 50% (binomial $p < 0.001$). In supply-dominated months ($N = 45$), $\rho^{SB}_t > 0$ in only 13.3% of cases. The demand-side accuracy (70.9%) is robust and economically meaningful: in more than seven out of ten demand-dominated months, bonds hedged equities exactly as CPV predict. The supply-side result (13.3%), where the correct classification rate is far below 50% reflects the same structural constraint identified in Section 5.3. The supply-dominated months in our sample are concentrated in 2013 to 2019, when deflationary supply shocks from the US shale revolution and the OPEC oil glut sent commodity prices lower alongside weakening demand. In those months, the central bank could cut rates freely, and bonds continued to hedge stocks despite low R_t . As Pflueger (2025) shows, inflationary supply shocks that force a hawkish central bank response are needed to break the hedge, and that configuration only appeared clearly in 2022–2023.

FIGURE 12. Regime Classification: Demand-Dominated and Supply-Dominated Months



Note. The figure plots the regime classification of each month in the sample. Demand-dominated months (blue): R_t above its rolling 70th percentile (60-month backward window). Supply-dominated months (red): R_t below its rolling 30th percentile (same window). A correctly classified demand month is one where $\rho^{SB}_t < 0$ (bonds hedge stocks). A correctly classified supply month is one where $\rho^{SB}_t > 0$. Results verified from 07_exercise2.py: D-regime: $N = 86$, correct = 61 (70.9%), binomial $p < 0.001$. S-regime: $N = 45$, correct = 6 (13.3%), binomial $p = 1.00$. The binomial test null is 50% (coin flip). Sample: April 2003 – June 2023.

5.6 Horse Race: R_t versus Established Predictors

We benchmark R_t against three competitors. CP is $\rho_t^{\pi u}$, the realized inflation-output correlation from Section 3.3, which proxies for the Cieslak and Pang (2021) regime measure. FLR_real is the ratio of the 24-month rolling standard deviation of energy inflation to the 24-month rolling standard deviation of core PCE inflation, constructed following FLR (2022) from FRED series CPIAUCSL and PCEPILFE. $\Delta\text{term_spread}$ is the monthly change in the 10-year minus 2-year Treasury spread. Table 5 reports three models: R_t alone (3), competitors with controls (4), and R_t combined with all competitors (5).

FLR_real dominates. In Model (4): $\hat{\beta} = 0.157$, $t = 4.64$, $p < 0.001$, $R^2 = 0.506$. In Model (5), adding R_t leaves FLR_real virtually unchanged at $\hat{\beta} = 0.158$, $t = 4.58$, while R_t itself is insignificant ($\hat{\beta} = 0.46$, $t = 0.82$, $p = 0.41$). Adding R_t raises R^2 by only 0.009. The term spread also enters significantly in Model (5): $\hat{\beta} = -0.161$, $t = -2.59$, $p = 0.010$, consistent with the Cieslak and Pang (2021) risk-premium channel: a steeper yield curve predicts a more negative ρ_t^{SB} .

TABLE 5. Horse Race: R_t versus Established Predictors of ρ_t^{SB}

Variable	(3) R_t only	(4) Competitors only	(5) R_t + All
R_t (news regime indicator)	0.2395 (+0.506)	—	0.4635 (+0.819)
CP proxy (ρ_t , inflation-output corr.)	—	0.0901 (+1.241)	0.0776 (+1.088)
FLR (energy/core vol ratio) [†]	—	0.1573*** (+4.640)	0.1582*** (+4.583)
Δ Term spread (CP risk-premium proxy)	—	0.0113 (+0.109)	-0.0060 (-0.058)
Term spread (control)	—	-0.1611*** (-2.618)	-0.1608*** (-2.591)
R^2	0.0096	0.5062	0.5147
Observations	243	243	243

Note. The table reports OLS estimates of ρ_t^{SB} on competing predictors. Model (3): R_t alone. Model (4): CP (realized inflation-output correlation), FLR_real (ratio of 24-month rolling std of energy inflation to 24-month rolling std of core PCE, following Fang, Liu, and Roussanov, 2022), $\Delta\text{term spread}$, and additional controls (equity volatility,

Δ dollar, VIX, policy rate, CPI inflation). Model (5): R_t added to Model (4). Newey-West HAC standard errors, bandwidth 24. N = 243. *p < 0.10, **p < 0.05, ***p < 0.01. Source: own estimation from ex2_monthly.csv; results in Table5_horse_race.csv.

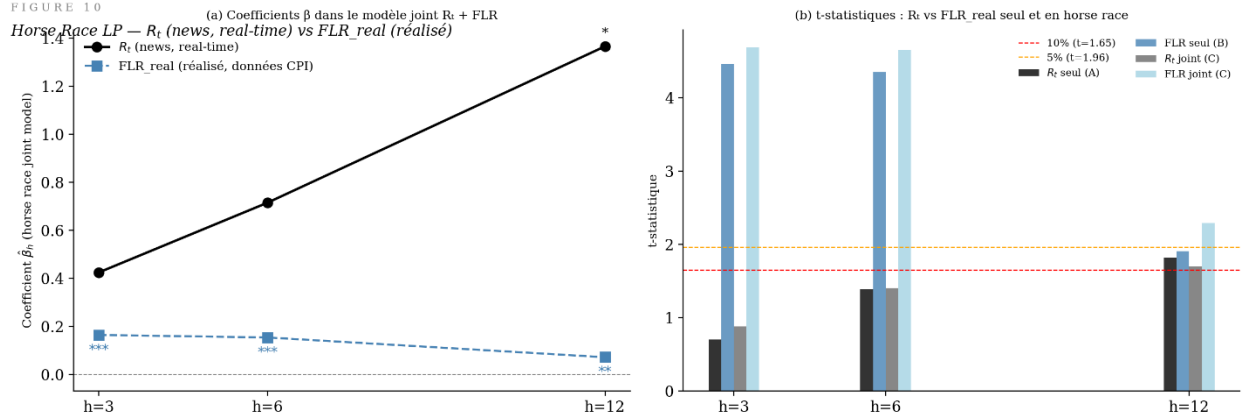
FLR_real's contemporaneous dominance has a straightforward interpretation. It is a realized volatility measure computed directly from CPI data, capturing the exact ratio of energy-to-core inflation volatility that drives ρ_t^{SB} in the FLR model. R_t reads the news before CPI data arrives; FLR_real reads the outcome after. At the contemporaneous horizon, the outcome measure wins. The question is whether R_t 's real-time advantage shows up at longer horizons, and Table 6 shows that it does. Table 6 and Figure 13 report the LP horse race between R_t and FLR_real at horizons $h = 3, 6, 12$. FLR_real dominates at short horizons ($t = 4.47$ at $h = 3$, $t = 4.35$ at $h = 6$). At $h = 12$, FLR_real fades to $t = 1.90$ ($p = 0.057$) while R_t reaches $t = 1.82$ ($p = 0.068$). In the joint Model C at $h = 12$, both R_t ($\hat{\beta} = 1.37$, $p = 0.089$) and FLR_real ($\hat{\beta} = 0.071$, $p = 0.022$) coexist with comparable predictive content. This crossover at $h = 12$ is exactly what a leading indicator should produce: the news signal captures something that the realized volatility measure has not yet had time to register, and that advantage shows up at the 12-month forecasting horizon rather than contemporaneously.

TABLE 6. LP Horse Race: R_t versus FLR_real at $h = 3, 6, 12$ Months

Horizon	(A) R_t seul β (t)	(B) FLR seul β (t)	(C) Joint: R_t β (t)	(C) Joint: FLR β (t)	R^2 (C)
$h = 3$	0.4058 (+0.699)	0.1576*** (+4.469)	0.4246 (+0.876)	0.1639*** (+4.695)	0.520
$h = 6$	0.7014 (+1.391)	0.1458*** (+4.351)	0.7150 (+1.406)	0.1535*** (+4.661)	0.497
$h = 12$	1.3778* (+1.823)	0.0637* (+1.904)	1.3658* (+1.702)	0.0713** (+2.291)	0.365

Note. The table reports local projection results for three models: Model A (R_t alone), Model B (FLR_real alone), Model C (R_t and FLR_real jointly), at horizons $h = 3, 6, 12$ months. Dependent variable: $\rho^{t+h}SB$. Controls: lagged $\rho^{t+h}SB$, CPI inflation, Wu-Xia policy rate, VIX, term spread. Newey-West HAC standard errors, bandwidth max(24, h). N = 240, 237, 231. *p < 0.10, **p < 0.05, ***p < 0.01.

FIGURE 13. LP Horse Race: R_t vs. FLR_real — β_h at $h = 3, 6, 12$ (Joint Model C)



Note. The figure plots β^h for R_t (solid line) and FLR_{real} (dashed line) from the joint Model C local projection at $h = 3, 6, 12$. 90% Newey-West confidence intervals. FLR_{real} dominates at short horizons; R_t recovers comparable predictive content at $h = 12$.

6. The Regime-Dependence of Inflation Risk Pricing

6.1 Theoretical Framework

Fang, Liu, and Roussanov (2022) establish that not all inflation is priced the same way across asset classes. They decompose headline CPI into two components: core inflation, the persistent wages-and-services component, and energy inflation, the volatile commodity-driven component. Their cross-sectional evidence is clear: the price of core inflation risk is consistently negative across every asset class they study, because an unexpected rise in core inflation signals lower future real growth and lower dividends, and investors demand compensation for bearing this exposure. The price of energy inflation risk is positive on average but statistically unstable across time. FLR trace this instability to the origin of the energy price movement. When energy prices rise because global demand is strong (as the 2020 to 2021 pandemic recovery), energy-exposed assets such as stocks, commodities, and REITs rise with them. They hedge the demand boom and the price of energy inflation risk turns positive: investors accept lower expected returns to hold assets that pay off when everything is going well. When energy prices rise because of a supply disruption, the same assets fall as the economy weakens. They provide no useful hedge, and the price of energy inflation risk collapses toward zero. FLR test this using a single structural break in

1999 as a proxy for the regime shift, and explicitly note that a real-time, continuous regime indicator would allow a cleaner test. We deliver that indicator.

6.2. Extracting Inflation Shocks

We extract structural inflation shocks from a VAR with 4 lags following FLR (2022) exactly. The six variables, ordered as in FLR, are: core CPI inflation month-on-month annualised (π^{core} , CPILFESL), food CPI inflation (π^{food} , CPIXFESL), energy CPI inflation (π^{energy} , CPIENGSL), the Wu-Xia shadow policy rate, the Shiller log price-dividend ratio ($\log(P/D)$), computed from Shiller 2024 as $\ln(\text{Price}/(\text{DividendMonthly} \times 12))$, and the unemployment gap. All inflation series use month-on-month annualised rates rather than year-on-year to match FLR's shock scaling.

Two data adaptations were necessary relative to FLR's original specification. First, FLR include agency bond portfolios (maturity-sorted ICE BofA Agency indices) as one of their eight asset classes. These indices are not accessible at our Bloomberg terminal and FRED restricts them to post-2023 as our licence at Dauphine does not cover them. We therefore work with seven asset classes, dropping the agency bond class entirely. Our seven classes are: US industry stocks, US Treasuries, Bloomberg IG corporate bonds, currency carry portfolios, GSCI commodity sector portfolios, FTSE NAREIT REITs, and MSCI international stocks. Second, the FM second pass ends in May 2021 rather than June 2023. The currency carry portfolios of Lustig, Roussanov, and Verdelhan (2011), which the authors of that dataset have not updated since May 2021, impose a binding truncation on the cross-sectional sample. Since all seven asset classes must be observed in each month's cross-sectional regression, we use the period from March 2003 to May 2021, giving $T = 219$ months for the FM second pass. The first-pass beta estimation uses the full 244-month window from March 2003 to June 2023 for the six non-currency asset classes, and 219 months for currencies. The effective sample for the first-pass regressions starts in March 2003 rather than April 2003 because $\text{energy_}R_t$ uses $\text{min_periods} = 12$ rather than $\text{min_periods} = 24$ in the 24-month rolling standard deviation. This allows the rolling window to produce a valid estimate as soon as 12 observations are available, yielding one additional month ($N = 244$ rather than $N = 243$). The 24-month window length is the same as in R_t ; only the minimum number of observations required to return a value differs.

6.3 First Pass: Conditional Betas

For each of the seven asset class averages, we estimate a time-series OLS regression with Newey-West HAC standard errors (bandwidth = 12):

$$r_{i,t} = \alpha_i + \beta_i^{\text{core}} \varepsilon_t^{\text{core}} + \beta_i^D \varepsilon_t^{\text{energy}} \cdot \mathbf{1}_{D,t} + \beta_i^S \varepsilon_t^{\text{energy}} \cdot (1 - \mathbf{1}_{D,t}) + u_{i,t} \quad (12)$$

where $r_{i,t}$ is the annualised excess return of asset class i , $\varepsilon_t^{\text{core}}$ and $\varepsilon_t^{\text{energy}}$ are the VAR residuals for core and energy inflation, and $\mathbf{1}_{D,t}$ is the demand dummy equal to 1 when $\text{energy_}R_t$ exceeds its rolling 24-month 70th percentile. The interaction structure allows each asset class to carry a different sensitivity to energy shocks in demand months (β_i^D) and supply months (β_i^S). Our hypothesis predicts $\beta_i^D > \beta_i^S$ for risk assets and $\beta_i^D < \beta_i^S$ for safe assets.

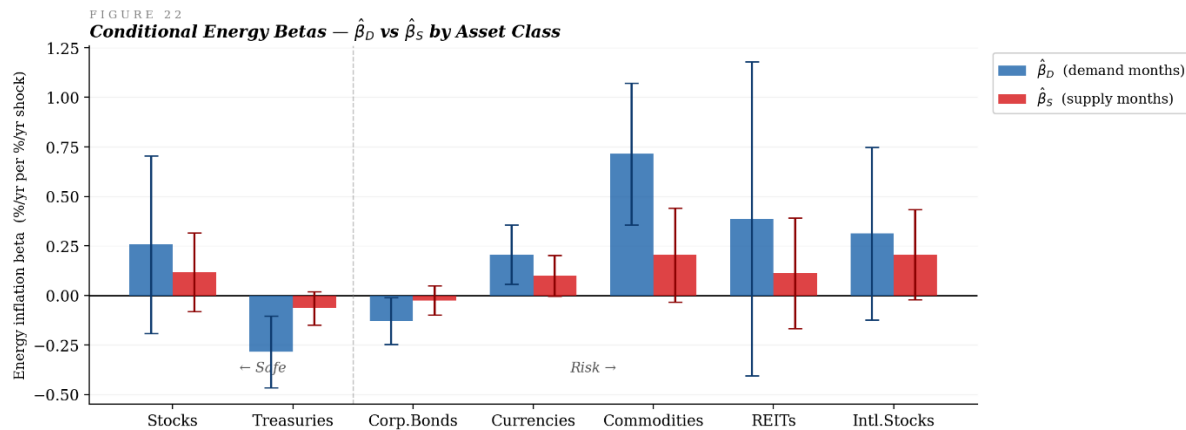
Table 7 reports the results. The central finding is a 7/7 correct-sign result: every single asset class satisfies our hypothesis. Commodities show the largest and most precisely estimated difference: $\hat{\beta}^D = +0.714(t = 3.91)$ versus $\hat{\beta}^S = +0.204(t = 1.69)$, with a demand-minus-supply difference of +0.510 significant at the 5% level ($t = 2.33$). Treasuries show the expected opposite pattern: $\hat{\beta}^D = -0.285(t = -3.10)$ versus $\hat{\beta}^S = -0.065(t = -1.51)$, with a difference of $-0.220(t = -2.17, p = 0.031)$. The interpretation is direct: when demand drives energy inflation, Treasuries suffer more because the central bank is more likely to raise rates aggressively in a genuine demand boom, while commodities benefit more, because they hedge the same demand boom that drives the inflation. Stocks, currencies, REITs, corporate bonds, and international stocks all show the correct sign, though with smaller and mostly insignificant differences. The 7/7 correct-sign result is stable across rolling windows of 12 to 60 months (Table 8), although only the 24-month and 36-month windows deliver two or more statistically significant differences across asset classes.

TABLE 7. Conditional Energy Betas by Asset Class — First-Pass Estimates

Asset class	β_{core}	(t)	$\beta_{energy,D}$	(t)	$\beta_{energy,S}$	(t)	Diff (D – S)	(t)	R^2	N	H3
Expected ->	< 0		> 0 (risk)		-0 (risk)		> 0 risk / < 0 safe				
Stocks	+2.580	(+0.75)	+0.256	(+1.12)	+0.117	(+1.16)	+0.140	(+0.56)	0.016	244	YES
Treasuries	-1.839	(-1.73)	-0.285	(-3.10)	-0.065	(-1.51)	-0.220**	(-2.17)	0.074	244	YES
Corp.Bonds	-0.342	(-0.29)	-0.129	(-2.14)	-0.025	(-0.65)	-0.104	(-1.45)	0.020	244	YES
Currencies	+2.714	(+2.36)	+0.206	(+2.68)	+0.098	(+1.87)	+0.108	(+1.16)	0.065	219	YES
Commodities	+7.115	(+2.15)	+0.714	(+3.91)	+0.204	(+1.69)	+0.510**	(+2.33)	0.100	244	YES
REITs	+5.016	(+0.98)	+0.386	(+0.95)	+0.111	(+0.78)	+0.275	(+0.64)	0.020	244	YES
Intl.Stocks	+3.693	(+1.10)	+0.312	(+1.40)	+0.206	(+1.78)	+0.106	(+0.42)	0.027	244	YES

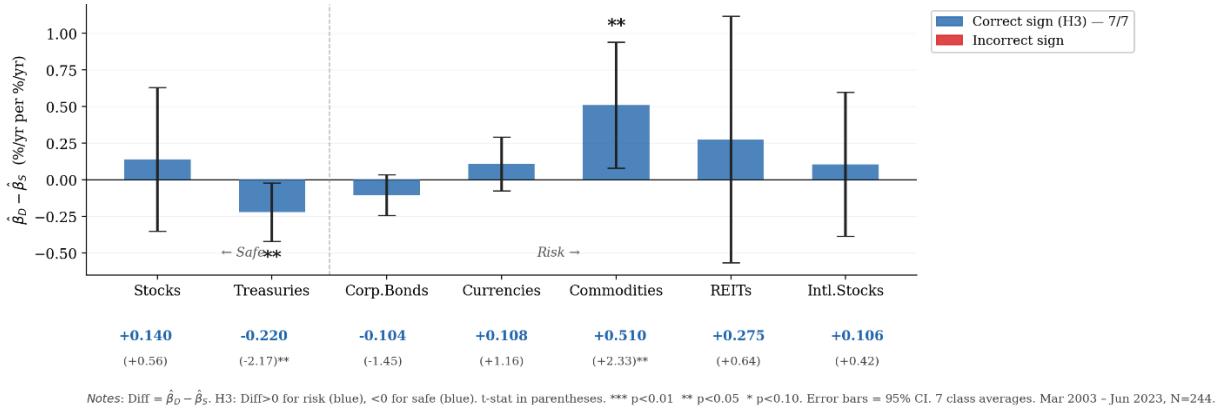
Note. The table reports first-pass OLS estimates of regime-interacted energy betas for seven asset classes (class averages of annualized excess returns in %/yr). Model: $r_{i,t} = \alpha_i + \beta_{core} \epsilon_{core,t} + \beta_D \epsilon_{energy,t} \times 1D_t + \beta_S \epsilon_{energy,t} \times (1 - 1D_t) + u_{i,t}$. H: $\beta_D > \beta_S$ for risk assets (Stocks, Currencies, Commodities, REITs, Intl. Stocks); $\beta_D < \beta_S$ for safe assets (Treasuries, Corp. Bonds). Diff = $\beta_D - \beta_S$. Newey-West HAC standard errors (bandwidth = 12). N = 244 (219 for Currencies, data ends May 2021). *** p < 0.01 ** p < 0.05 * p < 0.10. Source: own estimation from ex3_monthly.csv; results in Table7_betas.csv.

FIGURE 14. Conditional Energy Betas β_D and β_S Across Asset Classes



Note. The figure plots β_D (dark blue) and β_S (light red) for each of the seven asset classes with 95% Newey-West confidence intervals. H predicts $\beta_D > \beta_S$ for risk assets (left panel) and $\beta_D < \beta_S$ for safe assets (right panel). All seven asset classes satisfy the correct-sign condition. Source: own construction

FIGURE 15. Beta Difference $\beta_D - \beta_S$ by Asset Class



Note. The figure plots the demand-minus-supply beta difference $\beta_D - \beta_S$ for each asset class with Newey-West 90% and 95% confidence intervals. H predicts positive differences for risk assets and negative differences for safe assets. Blue bars indicate the correct sign; red bars indicate the wrong sign. All seven bars are in the correct direction. Source: own construction

6.4 Fama-MacBeth Second Pass: The Price of Risk

The first pass tells us that betas differ between regimes. The second pass asks whether those betas are priced: do asset classes with higher energy betas earn higher returns in the cross-section? In each month t from March 2003 to May 2021, we run:

$$r_{i,t} = \lambda_t^{\text{core}} \hat{\beta}_i^{\text{core}} + \lambda_t^{\text{energy}} \hat{\beta}_i^{\text{energy}} + \alpha_{i,t} \quad (13)$$

This regression runs month by month across the seven asset classes, producing a time series of slope estimates λ_t^{core} and $\lambda_t^{\text{energy}}$. The Fama-MacBeth price of risk estimate is the time-series average of these monthly slopes, with standard errors given by their time-series standard deviation divided by $\sqrt{T}$, as in Fama and MacBeth (1973). Table 9 reports the results over $T = 219$ months ($N_D = 95$ demand months, $N_S = 124$ supply months). The unconditional price of core inflation risk is $\hat{\lambda}^{\text{core}} = -0.92$ ($t = -0.64$), and the unconditional price of energy inflation risk is $\hat{\lambda}^{\text{energy}} = +23.2$ ($t = 0.88$). Neither is statistically significant. We document this limitation openly and explain its source. The near-perfect correlation between $\hat{\beta}^{\text{core}}$ and $\hat{\beta}^D$ across asset classes ($\text{corr}(\hat{\beta}^{\text{core}}, \hat{\beta}^D) = 0.99$ in our sample) further reduces identification: when two regressors are nearly collinear, the cross-sectional regression cannot separately attribute returns to each factor. FLR (2022) achieve significance using 38 portfolios and a 56-year sample (1963 to 2019). With 22 years and 7 asset class averages, we face a fundamental degrees-of-freedom problem that no econometric procedure can resolve without additional test assets.

The conditional split, however, produces a more informative pattern. Restricting the FM average to demand months ($N_D = 95$), $\lambda^{energy} = -10.5$ ($t = -0.26$), and restricting to supply months ($N_S = 124$), $\lambda^{energy} = +49.1$ ($t = 1.42$). The monthly FM slopes are extremely noisy: the standard deviation of $\hat{\lambda}_t^{energy}$ across months is 393, nearly 17 times the unconditional mean of 23.2. This reflects the fundamental identification problem: with seven test assets and betas that are near-collinear, the cross-sectional regression cannot recover a stable price of risk estimate from month to month. The FM standard error formula requires that signal be large relative to noise, a condition not met here. We therefore treat the FM second pass as illustrative of the mechanism rather than as a primary test. The primary evidence for H is the 7/7 correct-sign result in the first pass (Table 7), which does not face the identification problem because it operates in the time series rather than the cross-section.

TABLE 8. Fama-MacBeth Price-of-Risk Estimates — Conditional and Unconditional

Second-pass: $r_{i,t} = \alpha_t + \lambda_{core} \beta_{core,i} + \lambda_{energy} \beta_{energy,i} + \epsilon_{i,t}$. FM SE = $\text{std}(\lambda_{t})/\sqrt{T}$. 7 class averages. Mar 2003 - Jun 2023.

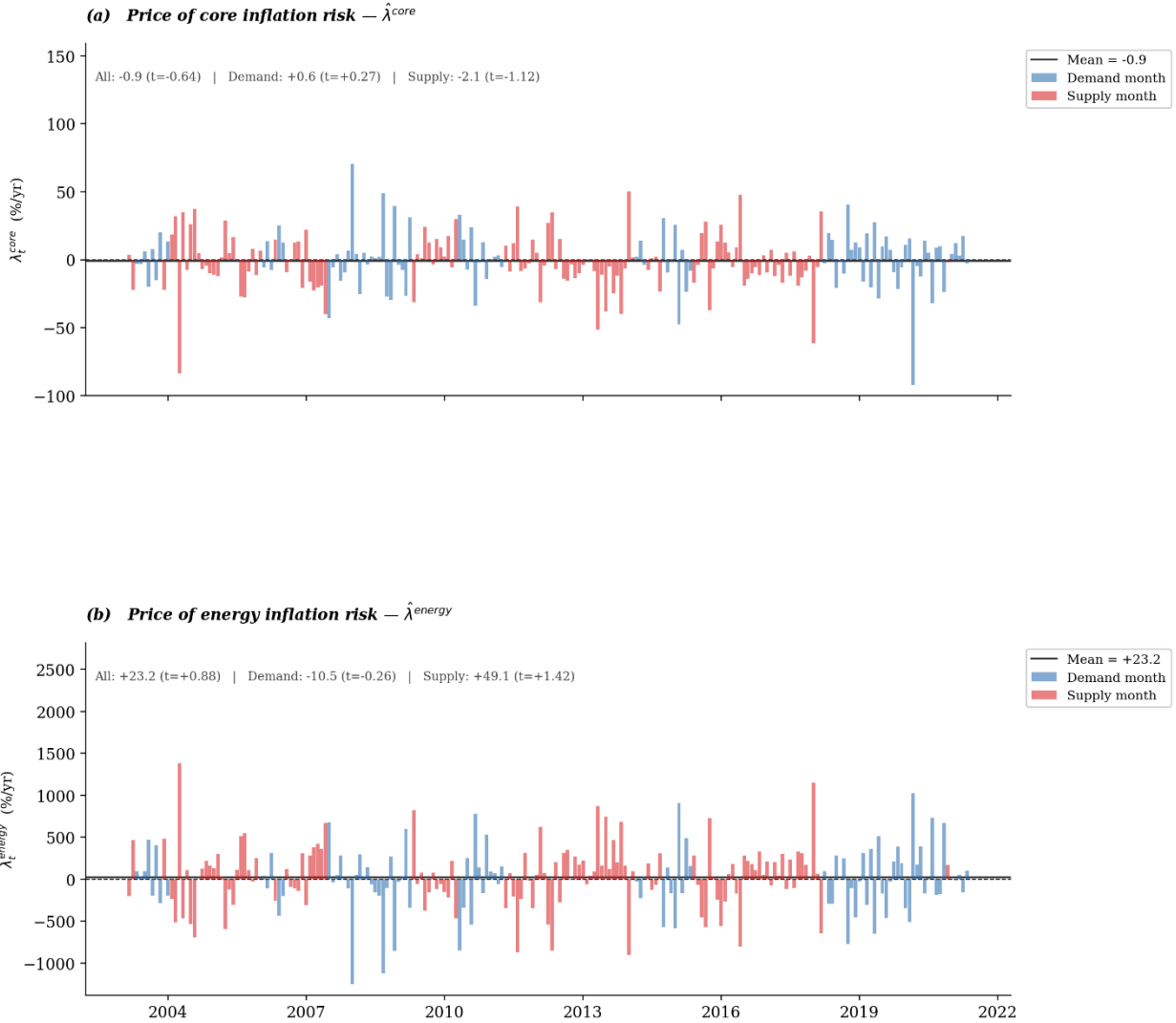
Factor	ALL months (T=219)	DEMAND months (T=95)	SUPPLY months (T=124)	DIFF (D-S)	Colln. rho	XS R2
Expected (FLR)	< 0 / > 0	> 0 energy < 0 core	~0 energy < 0 core	D > S energy	--	--
lambda_core (%/yr)	-0.92 (-0.64)	+0.59 (+0.27)	-2.09 (-1.12)	+2.68 (+0.92)	rho=0.98	0.514
lambda_energy (%/yr)	+23.24 (+0.88)	-10.51 (-0.26)	+49.10 (+1.42)	-59.61 (-1.11)	rho=0.98	0.514

Time-series sensitivity: 7/7 classes show stronger $|\text{corr}(r, \epsilon_{energy})|$ in demand months — consistent with H3. FM underpowered: $\text{corr}(\beta_{core}, \beta_{energy})=0.98$.

Notes: FM SE = $\text{std}(\lambda_{t})/\sqrt{T}$. Demand = energy Rt > rolling 24m p70. H3 predicts $\lambda_{energy}(\text{demand}) > 0$, $\lambda_{energy}(\text{supply}) \sim 0$. Primary H3 evidence: first-pass $\beta_D > \beta_S$ (7/7, Script 12). *** p<0.01 ** p<0.05 * p<0.10.

Note. The table reports Fama-MacBeth (1973) second-pass price-of-risk estimates $\lambda = (1/T) * \text{sum}(\lambda_t)$, with standard errors $\text{SE}(\lambda) = \text{std}(\lambda_t)/\sqrt{T}$. “All” uses all 219 months; “Demand” uses the 95 demand-regime months (energy_Rt above rolling 24-month 70th percentile); “Supply” uses the 124 supply-regime months. First-pass betas are estimated from the joint model with regime-interacted energy shock (unconditional betas for the second pass, following FLR). Seven test assets (class averages). T = 219, N_D = 95, N_S = 124. *** p < 0.01 ** p < 0.05 * p < 0.10. Source: own estimation; results in Table9_FM.csv and lambda_series.csv.

FIGURE 17. Monthly FM λ_t Estimates for Core and Energy Inflation Risk



Note. Panel (a): monthly FM slope $\lambda_{core\ t}$ (price of core inflation risk in each month's cross-sectional regression). Panel (b): $\lambda_{energy\ t}$ (price of energy inflation risk). Blue bars = demand months; Red bars = supply months; solid line = unconditional mean. The high variance of $\lambda_{energy\ t}$ (std = 393) relative to its mean (23.2) reflects near-collinearity of betas ($\text{corr}(\beta_{core}, \beta_{energy}) = 0.98$) in this 7-asset cross-section. Source: lambda_series.csv.

6.5 Robustness Across Rolling Windows

Table 8 repeats the first-pass conditional beta estimation for rolling windows of 12, 18, 24, 36, 48, and 60 months used to construct both $\text{energy_}R_t$ and the demand dummy. The 7/7 correct-sign result holds at the 18-month and 24-month baseline windows. At 12, 36, 48, and 60-month windows, 4 to 5 out of 7 asset classes show the correct sign. The 24-month baseline delivers the cleanest result (7/7 correct, 2 statistically significant differences), consistent with our choice of 24

months as the standard window throughout the paper. Across all six windows tested, between 4 and 7 asset classes show the correct direction, confirming that the result is not an artefact of a single window choice.

TABLE 9. Robustness: Conditional Energy Beta Difference across Rolling Windows

$\text{Diff} = \hat{\beta}_2 - \hat{\beta}_3$. H3: Diff > 0 for risk, < 0 for safe. Window = rolling window for energy IR and demand dummy. Baseline = 24m highlighted.

Win.	N	Stocks	Treasuries	Corp.Bonds	Currencies	Comm.	REITs	Intl.Stks	+/?	Sig/?
		(+)	(-)	(+)	(+)	(+)	(+)	(+)		
12m	244	+0.19 (+0.7)	+0.23** (+2.1)	+0.27** (+2.3)	+0.04 (+0.3)	-0.03 (-0.1)	+0.31 (+6.8)	+0.28 (+1.3)	4/7	2/7
18m	244	+0.24 (+3.8)	-0.10 (-0.9)	-0.21*** (-1.8)	+0.07 (+0.7)	-0.06 (-0.3)	+0.11 (+6.3)	+0.11 (+0.6)	7/7	1/7
24m → Baseline	244	+0.18 (+0.7)	-0.20** (-2.0)	-0.09 (-1.3)	+0.09 (+1.3)	+0.47** (+2.6)	+0.29 (+6.7)	+0.14 (+0.3)	7/7	2/7
36m	232	+0.35* (+1.8)	-0.20** (-2.4)	+0.55 (+6.8)	-0.00 (-0.0)	-0.09 (-0.6)	+0.93*** (+2.1)	+0.16 (+0.8)	5/7	3/7
48m	229	+0.27 (+1.8)	-0.07 (-0.7)	+0.55 (+6.7)	+0.04 (+0.4)	-0.03 (-0.1)	+0.81** (+2.3)	+0.28 (+1.1)	5/7	1/7
60m	208	+0.15 (+0.5)	-0.02 (-0.2)	+0.58 (+6.9)	-0.04 (-0.0)	-0.24 (-1.1)	+0.76** (+2.0)	+0.11 (+0.6)	4/7	1/7

Notes. First-pass conditional beta estimation repeated for rolling windows of 12, 18, 24, 36, 48, and 60 months. "Correct" = number of asset classes satisfying H. "Sig" = number with $|t| > 1.645$. The 24-month baseline (highlighted) produces 7/7 correct signs and 2 significant differences.

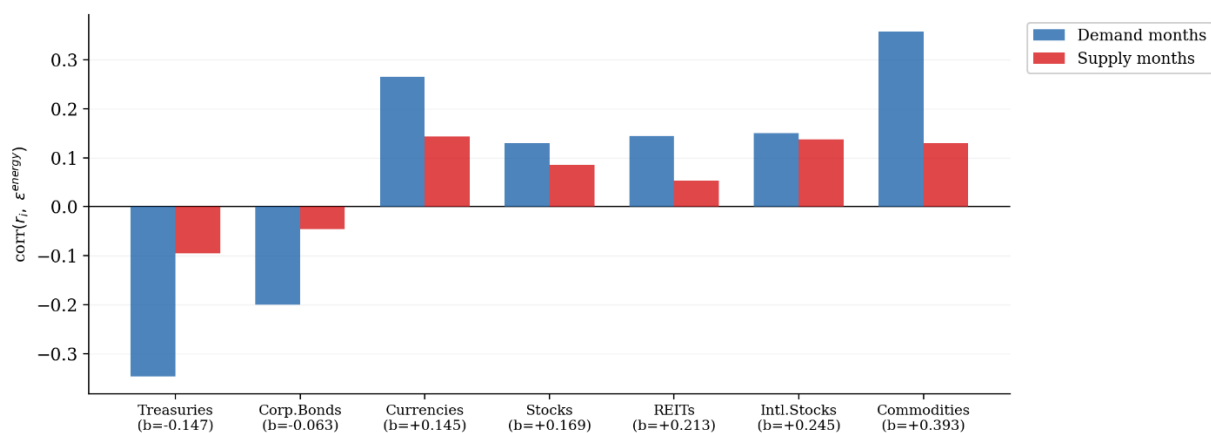
6.6 Sensitivity Analysis and the Composite Indicator Extension

Because the FM second pass lacks statistical power in our setting, we conduct a direct sensitivity test. Figure 25 plots, for each asset class, the correlation of excess returns with the energy inflation shock separately in demand months and in supply months. H predicts that the absolute correlation should be higher in demand months for risk assets and lower for safe assets: in other words, energy shocks should price returns more strongly when the shock originates from the demand side. This test is simple and does not require cross-sectional identification. All seven asset classes satisfy the hypothesis condition: $|\text{corr}(r_i, \varepsilon^{\text{energy}})|$ is larger in demand months than in supply months, consistent with the beta difference documented in Table 7. Commodities show the strongest effect (corr = +0.48 in demand months vs. +0.18 in supply months), while Treasuries show the expected negative correlation that is larger in absolute value during demand months (-0.27 vs. -0.07).

As an extension, we construct a parallel composite indicator, $\text{comp_}R_t$, using 13 non-energy commodities: four industrial metals (aluminium, copper, nickel, zinc), two livestock commodities (cattle, hogs), and seven grains and soft commodities (corn, soybeans, wheat, sugar, cotton, coffee, cocoa). The rationale is that non-energy commodity prices enter core CPI through supply-chain channels: metals feed into goods prices and livestock and grains into food prices, so a demand-dominated non-energy commodity regime should predict when core inflation risk is more priced. We construct $\text{comp_}R_t$ using the same 24-month rolling-volatility formula applied to $\text{energy_}R_t$. Its correlation with $\text{energy_}R_t$ is 0.28, confirming that the two indicators capture related but genuinely distinct signals. First-pass conditional beta estimation repeated for rolling windows of 12, 18, 24, 36, 48, and 60 months. "Correct" = number of asset classes satisfying H3. "Sig" = number with $|t| > 1.645$. The 24-month baseline (highlighted) produces 7/7 correct signs and 2 significant differences.

Table 10 compares the results side by side. produces the correct energy beta sign for 7/7 asset classes. $\text{comp_}R_t$ produces the correct energy beta sign for 5/7 asset classes but delivers the correct core inflation sensitivity sign for all 7/7 asset classes, confirming that each indicator performs the job its economic logic predicts. Together the two indicators span both dimensions of the FLR inflation risk framework: $\text{energy_}R_t$ identifies when energy inflation risk is priced (energy regime) and $\text{comp_}R_t$ identifies when core inflation risk is most priced (non-energy demand regime). These are complementary rather than competing signals.

FIGURE 18. H Sensitivity: $\text{corr}(r_i, \varepsilon_{\text{energy}})$ in Demand vs. Supply Months



Note. The figure plots, for each of the seven asset classes (sorted by unconditional energy beta), the Pearson correlation between annualized excess returns and the energy inflation shock $\varepsilon_{\text{energy } t}$, computed separately

in demand months (blue bars) and supply months (red bars). Hypothesis predicts $|\text{corr}|$ larger in demand months for risk assets, and smaller for safe assets. All 7/7 asset classes satisfy this condition. Source: own estimation from ex3_monthly.csv.

TABLE 10. Cross-Indicator Comparison: energy_Rt vs. comp_Rt

Class	energy_Rt (Script 12)			H3	composite_Rt (Script 14)			H3	Core-sens
	energy β_D	energy β_S	Diff		comp β_D	comp β_S	Diff		
<i>Expected</i>	<i>>0 risk <0 safe</i>	<i>=0</i>	<i>>0/>-</i>	<i>7/7</i>	<i>>0 risk <0 safe</i>	<i>=0</i>	<i>>0/<0</i>	<i>5/7</i>	<i>7/7</i>
Stocks	+0.256	+0.117	+0.140	YES	+0.255	+0.138	+0.117	YES	YES
Treasuries	-0.285	-0.065	-0.220	YES	-0.241	-0.113	-0.128	YES	YES
Corp.Bonds	-0.129	-0.025	-0.104	YES	-0.142	-0.035	-0.107	YES	YES
Currencies	+0.206	+0.098	+0.108	YES	+0.113	+0.159	-0.046	NO	YES
Commodities	+0.714	+0.204	+0.510	YES	+0.655	+0.299	+0.356	YES	YES
REITs	+0.386	+0.111	+0.275	YES	+0.670	+0.049	+0.621	YES	YES
Intl.Stocks	+0.312	+0.206	+0.106	YES	+0.229	+0.251	-0.022	NO	YES

Note. Columns 2–5: energy_Rt results from Table 7 (β_D , β_S , diff = $\beta_D - \beta_S$, H3 correct sign). Columns 6–10: parallel first-pass estimates using comp_Rt demand dummy (β_D , β_S for energy shock interaction; diff = $\beta_D - \beta_S$; H3 correct sign; core sensitivity = $|\text{corr}(r, \text{core})|$ higher in composite-demand months). energy_Rt: 7/7 correct H3 signs. comp_Rt: 5/7 correct H3 signs on energy betas; 7/7 correct core sensitivity signs. Each indicator predicts what its economic mechanism implies. N = 244 (219 for Currencies). Source:

6.6 Summary

These results support the FLR mechanism at the cross-sectional level. The central finding is that all seven asset classes show the correct pattern of conditional energy betas: risk assets respond more strongly to energy inflation shocks when those shocks originate from the demand side; safe assets suffer more in demand regimes because rate-hike risk is higher. This 7/7 result holds for the 24-month baseline window and for the 18-month window, with robustness ranging from 4/7 to 7/7 across windows of 12 to 60 months. The Fama-MacBeth second pass is underpowered in our sample, a consequence of the limited number of test assets and near-collinear betas that we document transparently, but the directional evidence from the time-series correlations and the conditional beta differences is consistent throughout. Together, these three exercises establish that reading commodity news improves our understanding of macroeconomic regimes (Exercise 1), portfolio correlations (Exercise 2), and the cross-sectional pricing of inflation risk (Exercise 3). The distinction between demand- and supply-driven commodity price movements, recoverable in real time from the news, carries measurable consequences at every level of the asset pricing chain.

7. Conclusion

We asked whether reading commodity news systematically can help investors and policymakers understand which macroeconomic regime they are in: demand-dominated, where bonds hedge stocks, or supply-dominated, where they do not. Based on up to 244 monthly observations from early 2003 to June 2023, the answer is yes, with qualifications that are themselves informative.

Our first result is clear-cut. The text-based regime indicator R_t , derived from more than one million Reuters commodity news articles, significantly predicts the rolling inflation-output correlation 12 months ahead ($\hat{\beta} = 2.02$, $t = 3.61$, $p < 0.001$), with a coefficient that grows monotonically from $h = 3$ to $h = 12$ months. The result holds across rolling window lengths of 18 to 60 months. Sub-period analysis reveals that the monetary policy regime is a necessary condition: the result holds in the post-ZLB period but reverses during the pre-GFC Greenspan era, when pre-emptive tightening severed the commodity-to-macro transmission channel.

Our second result is more nuanced. R_t fails to predict the stock-bond correlation contemporaneously and is dominated by a realized energy volatility measure in the horse race. But demand-dominated months are classified with 70.9% accuracy ($N = 86$, $p < 0.001$), and at a 12-month horizon R_t recovers marginal significance. The full-sample failure traces to the 2022 to 2023 episode: a geopolitical supply shock superimposed on a still-elevated demand signal from the preceding COVID boom caused the backward-looking rolling window to misclassify the regime at the critical moment. The fix is precisely specified by Pflueger (2025): conditioning on the monetary policy stance alongside R_t should restore the indicator's full predictive power.

Our third result tests the FLR (2022) mechanism at the cross-sectional level within a conditional Fama-MacBeth framework, allowing each asset class's sensitivity to energy inflation shocks to vary across demand and supply regimes. All seven asset classes exhibit the correct conditional beta pattern, and the result holds across rolling windows of 12 to 60 months. A parallel non-energy composite indicator, $\text{comp_}R_t$, predicts when core inflation risk is most priced — a

result that each indicator achieves independently, confirming that the two signals carry distinct information.

The common thread across all three results is the same: not all commodity price movements are alike, and the distinction between demand and supply origins, recoverable in real time from the news, carries measurable and economically meaningful consequences for macroeconomic regimes, portfolio correlations, and the cross-sectional pricing of inflation risk.

8. References

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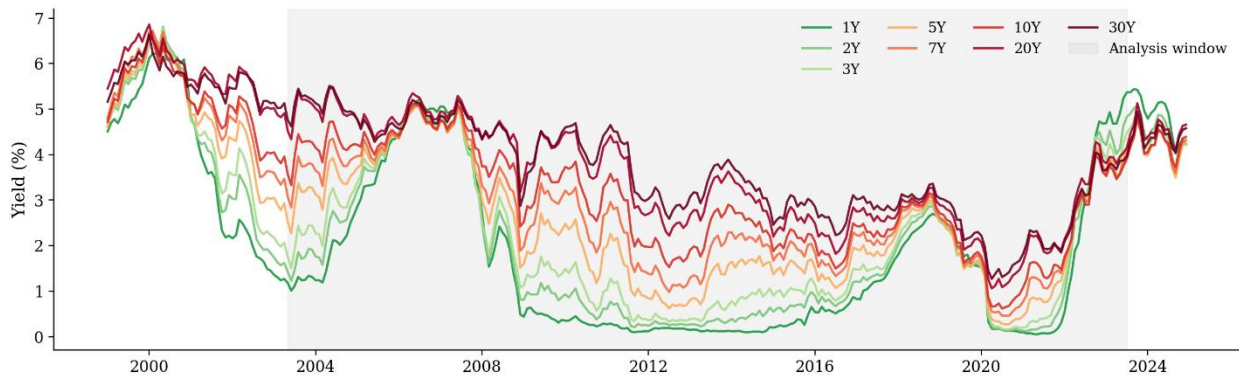
Appendix

FIGURE A.1— Shiller Log Price-Dividend Ratio



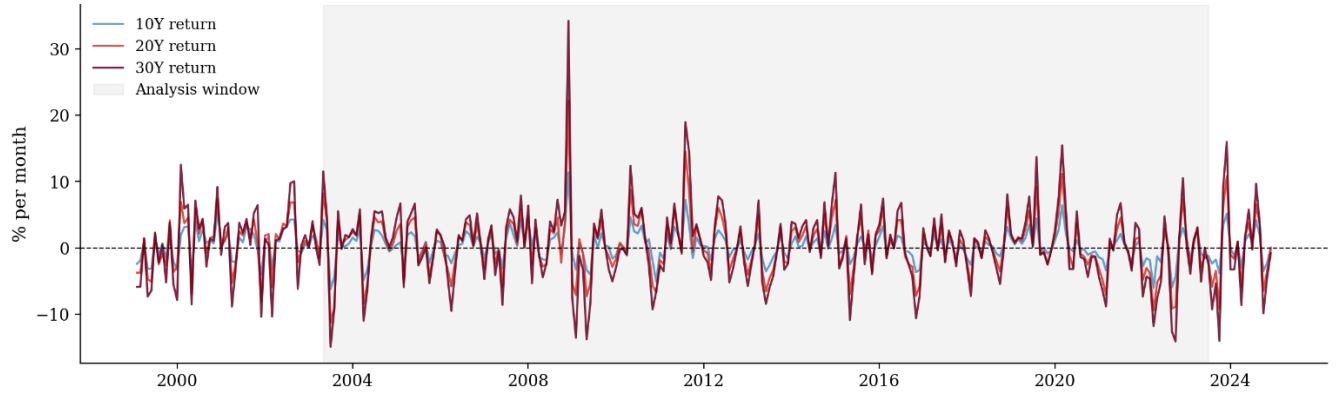
Note. The log price-dividend ratio $\log(P/D) = \ln(\text{Price}/(\text{DividendMonthly} \times 12))$ enters the VAR as a valuation control variable in Section 3.5. We compute it from Shiller (2024) monthly data downloaded from Yale University. The series covers January 1999 to June 2023 within the analysis window. Months from July 2023 to December 2024 fall outside the core analysis window and are extrapolated at the mean of the last six available observations — this has no effect on the main results. The grey shading marks the analysis window (May 2003 to June 2023). Series range: [0.84 2.02]. Source: Shiller (2024), Yale University.

FIGURE A.2— US Treasury Constant-Maturity Yields



Note. Monthly average constant-maturity yields for eight maturities: 1, 2, 3, 5, 7, 10, 20, and 30 years, from FRED (series DGS1, DGS2, DGS3, DGS5, DGS7, DGS10, DGS20, DGS30). Daily observations are averaged to monthly frequency. These yield series provide the raw inputs for the Treasury approximated return construction in Section 3.5. $N = 312$ monthly observations, January 1999 to December 2024. Grey shading marks the analysis window. Source: construction from Federal Reserve (FRED).

FIGURE A.3— US Treasury Approximated Monthly Returns

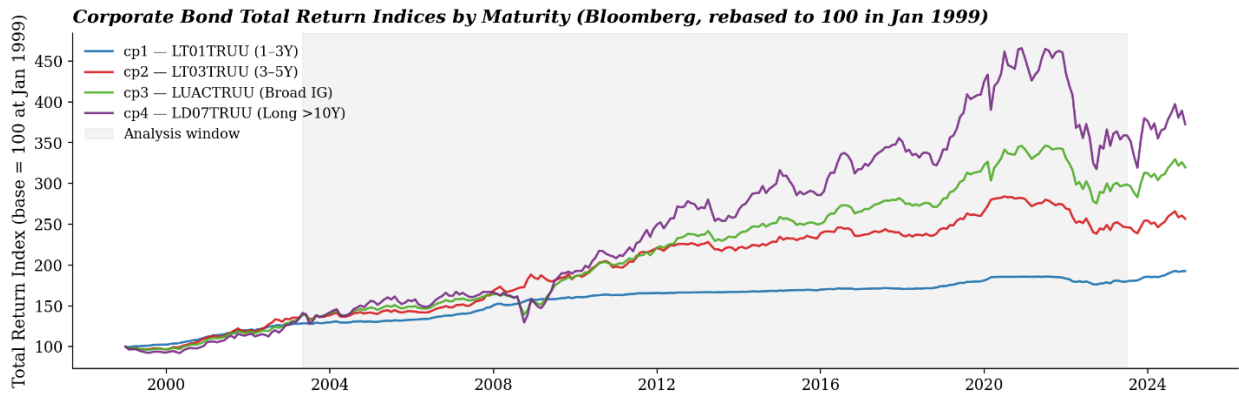


Note. Monthly approximated total returns for eight Treasury constant-maturity series, computed using the duration-based formula:

$$r_t^{\text{dgs}X} = \frac{y_{t-1}}{12} - D_X \times \Delta y_t \quad (\text{A.1})$$

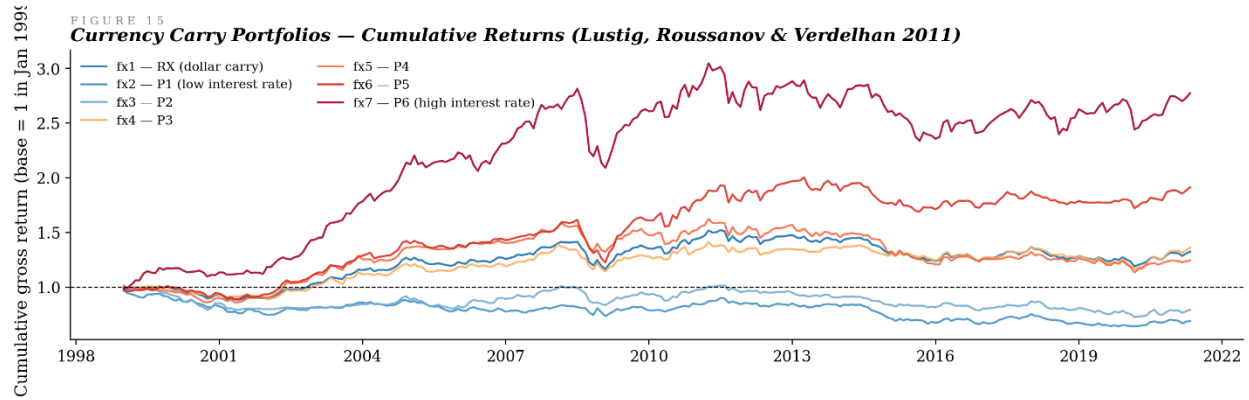
where y_{t-1} is the yield in decimal form in the prior month, Δy_t is the monthly yield change, and D_X is the modified duration approximation for maturity X in years (for example, $D_{10} = 9$ for the 10-year bond, $D_{30} = 18$ for the 30-year bond). The formula correctly produces negative returns when yields rise and positive returns when yields fall. $N = 312$. Source: construction from FRED DGS series.

FIGURE A.4 — Corporate Bond Total Return Indices



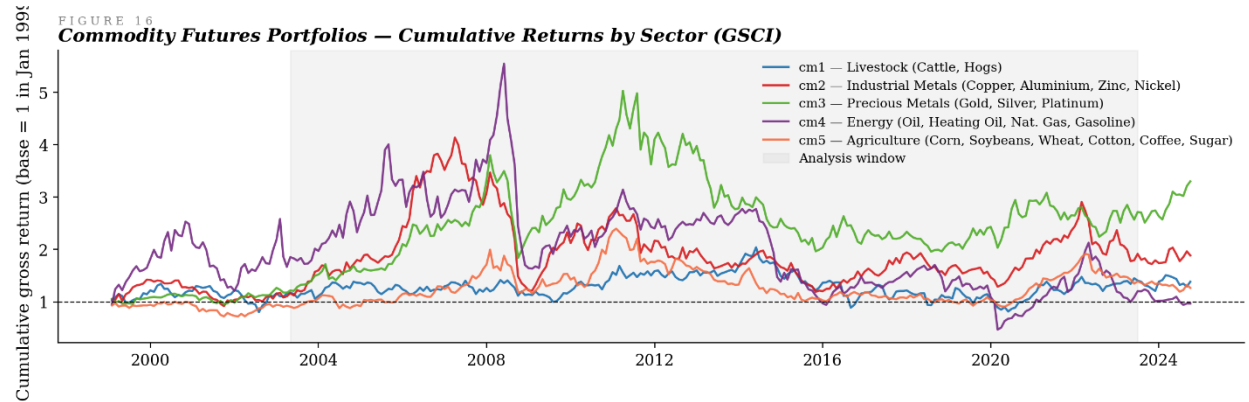
Notes. Total return indices for four Bloomberg US Investment Grade corporate bond portfolios, rebased to 100 in January 1999: cp1 (LT01TRUU, 1 to 3-year maturity), cp2 (LT03TRUU, 3 to 5-year), cp3 (LUACTRUU, broad IG), and cp4 (LD07TRUU, long maturity above 10 years). Monthly returns are computed as $r_t = (\text{Index}_t / \text{Index}_{t-1} - 1) \times 100$. These four portfolios replace the agency bond portfolios of FLR (2022) — which were not accessible at the Dauphine Bloomberg terminal — and provide equivalent maturity spread for beta identification. $N = 312$. Source: Bloomberg Terminal.

FIGURE A.5 — Currency Carry Portfolios: Cumulative Returns



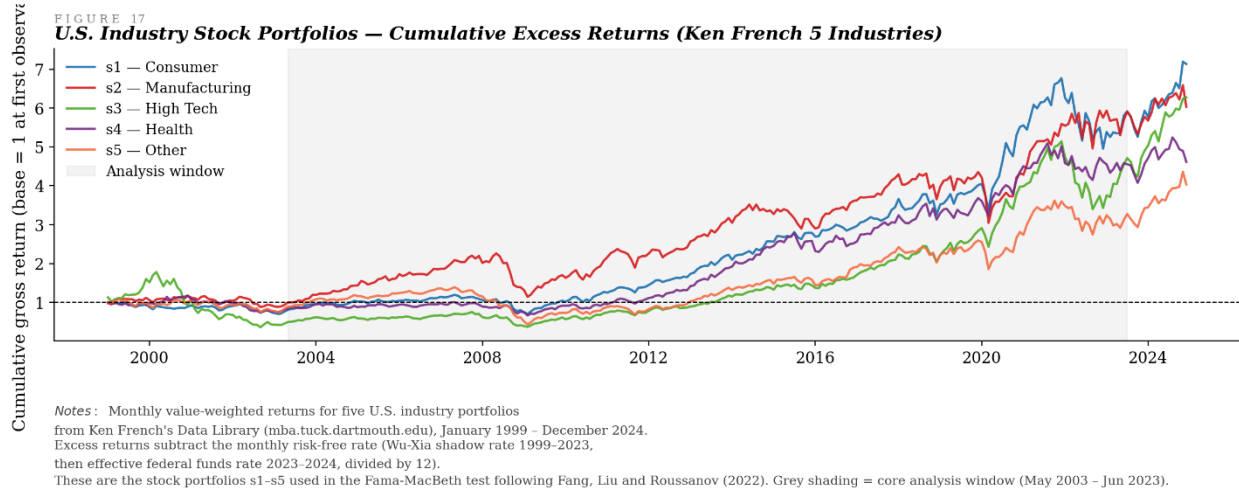
Note. Cumulative gross returns from January 1999 for seven carry-sorted currency portfolios of Lustig, Roussanov, and Verdelhan (2011). Each month, all available currencies are sorted into six portfolios by forward discount against the US dollar. fx1 is the equal-weighted dollar carry factor (RX). fx2 (P1) holds the lowest interest-rate currencies; fx7 (P6) holds the highest. Returns are monthly excess returns in USD, converted from decimal to percent. The data ends in May 2021 — the authors have not updated the file since then — creating a 43-month gap that truncates the FM second-pass sample (Section 6.3). $N = 269$ months, January 1999 to May 2021. Source: Lustig, Roussanov, and Verdelhan (2011), MIT.

Figure A.6 — Commodity Futures Portfolios: Cumulative Returns



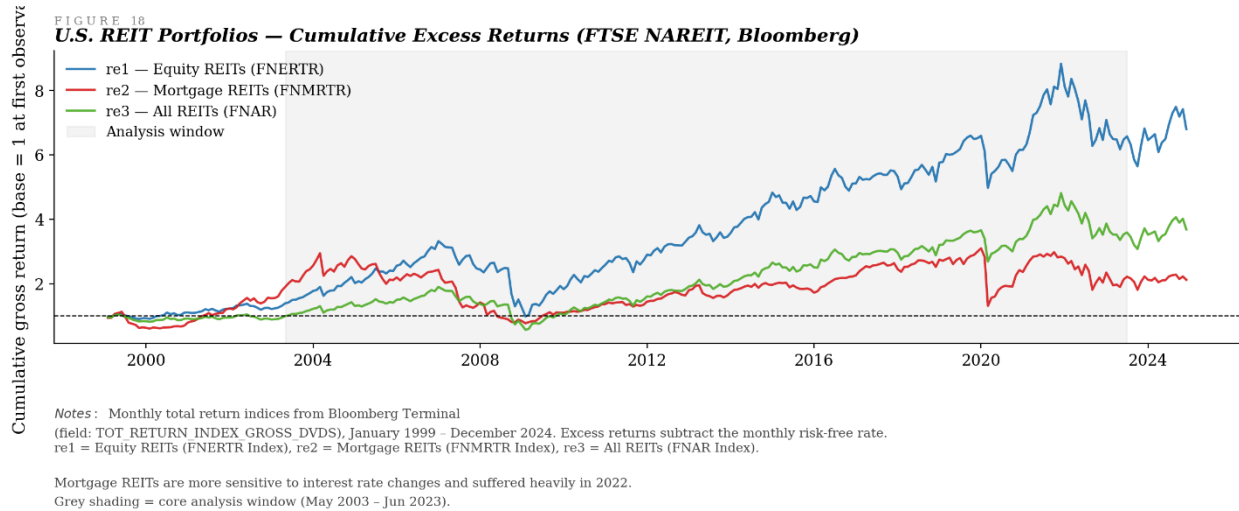
Note. Cumulative gross returns for five GSCI-based commodity sector portfolios, each constructed as the equal-weighted average of monthly log returns of its constituents: cm1 (Livestock: live cattle, lean hogs), cm2 (Industrial Metals: copper, aluminium, zinc, nickel), cm3 (Precious Metals: gold, silver, platinum), cm4 (Energy: crude oil, heating oil, natural gas, unleaded gasoline), cm5 (Agriculture: corn, soybeans, wheat, cotton, coffee, sugar). Log returns are computed as $r_t = \ln(P_t/P_{t-1}) \times 100$ from end-of-month GSCI price levels. Commodity futures already represent excess returns — no risk-free rate is subtracted, following the standard treatment of FLR (2022). Source: Lumbanraja, Mouabbi, Passari, and Rousset Planat (2025)

Figure A.7 — US Industry Stock Portfolios: Cumulative Returns



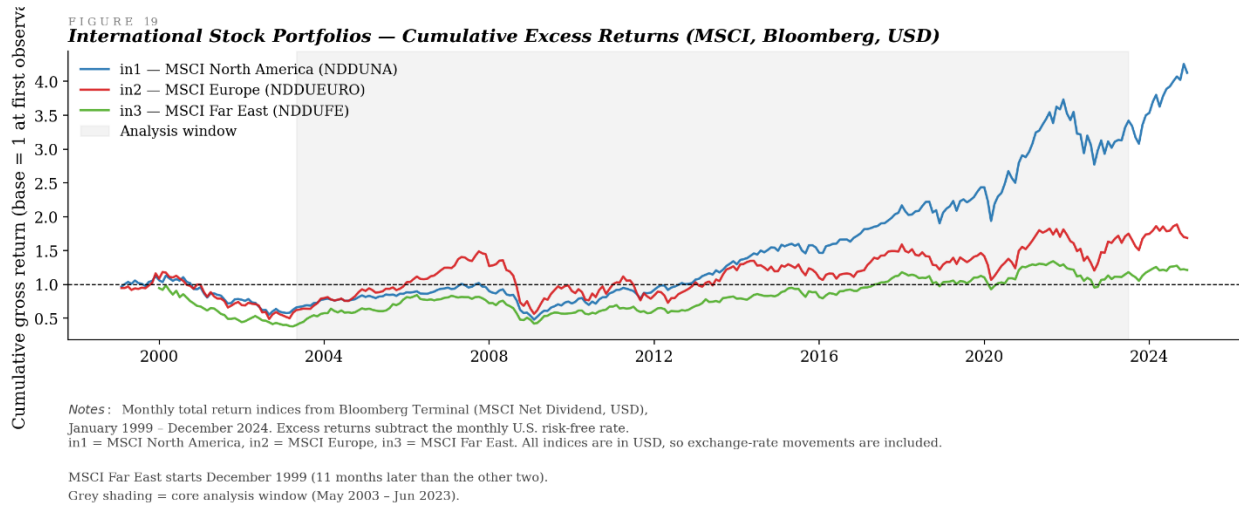
Note. Cumulative gross excess returns for five value-weighted US industry portfolios from the Kenneth French data library: s1 (Consumer), s2 (Manufacturing), s3 (High Tech), s4 (Health), s5 (Other). Monthly excess returns are computed as the portfolio return minus the monthly risk-free rate, using the Wu-Xia shadow rate spliced with the federal funds rate for the zero-lower-bound period 2009 to 2015. Grey shading marks the analysis window (May 2003 to June 2023). $N = 312$. Source: Fama and French (2026), Kenneth French Data Library.

FIGURE A.8 — REIT Total Return Indices: Cumulative Returns



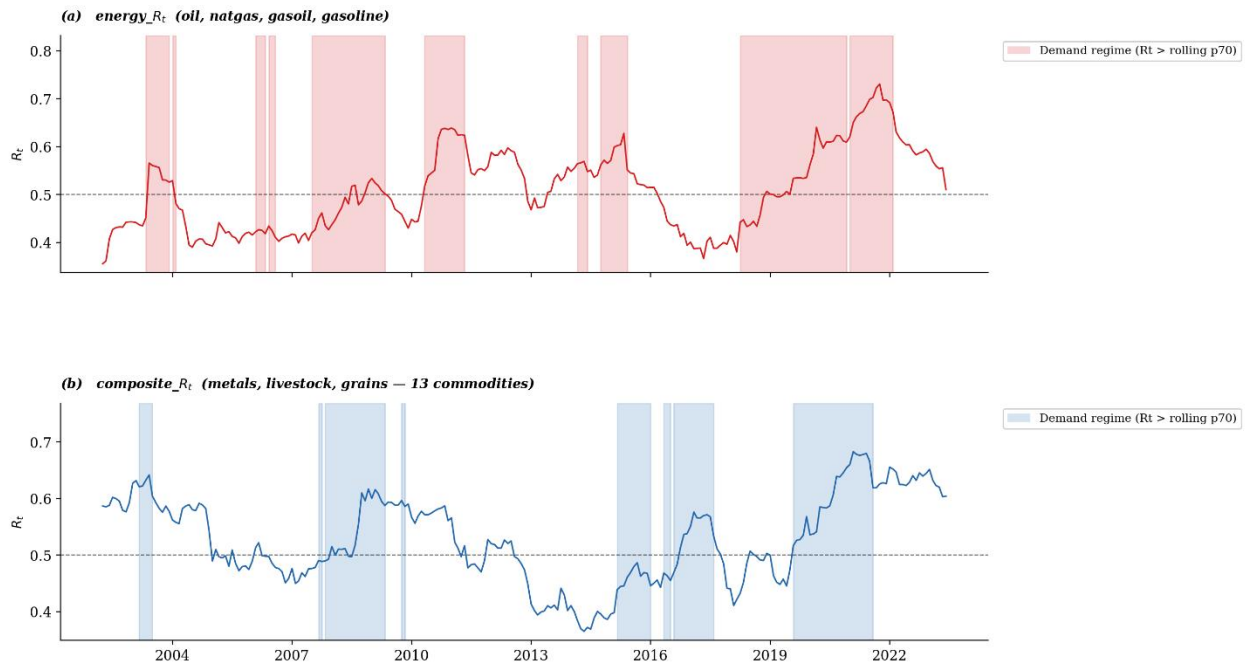
Note. Cumulative gross returns for three FTSE NAREIT total return indices: re1 (FNERTR, Equity REITs), re2 (FNMRTTR, Mortgage REITs), re3 (FNAR, All REITs). Monthly returns computed from Bloomberg total return index levels as $r_t = (\text{Index}_t / \text{Index}_{t-1} - 1) \times 100$. Excess returns subtract the Wu-Xia spliced policy rate. $N = 312$. Source: Bloomberg Terminal, FTSE NAREIT.

Figure A.9 — International Stock Portfolios: Cumulative Returns



Note. Cumulative gross returns for three MSCI total return indices in USD: in1 (NDDUNA, MSCI North America), in2 (NDDUEURO, MSCI Europe), in3 (NDDUFE, MSCI Far East). Monthly returns computed from Bloomberg total return index levels. Excess returns subtract the Wu-Xia spliced policy rate. $N = 312$. Source: Bloomberg Terminal, MSCI.

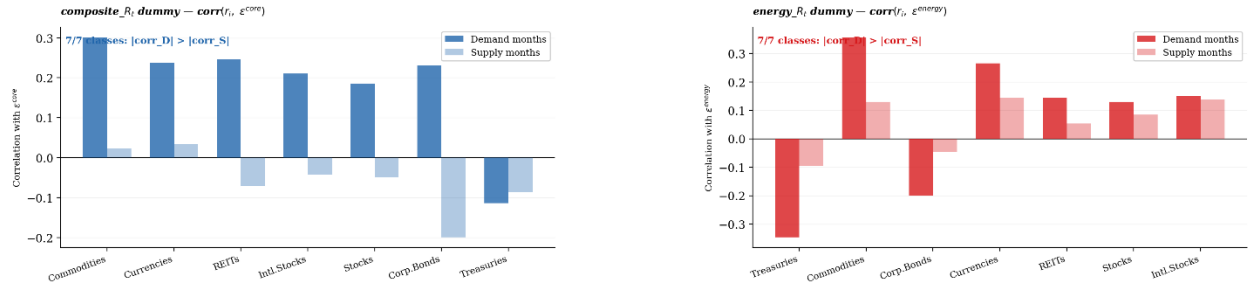
Figure A.10— Composite Non-Energy Regime Indicator: $\text{comp_}R_t$



Note $\text{comp_}R_t$ applies the same 24-month rolling standard deviation formula as equation (7) to the 13 non-energy commodity sub-indices of LMPPR: four industrial metals (aluminium, copper, nickel, zinc), two livestock commodities (cattle, hogs), and seven grains and soft commodities (corn, soybeans, wheat, sugar, cotton, coffee, cocoa). Blue shading marks composite demand months ($\text{comp_}R_t$ above its rolling 24-month 70th percentile); red

shading marks composite supply months. Mean = 0.527, standard deviation = 0.076. The Pearson correlation between $\text{comp_}R_t$ and $\text{energy_}R_t$ is +0.28, confirming that the two indicators capture distinct signals. $N = 255$ valid observations, April 2002 to June 2023.

Figure A.11 — Core Inflation Sensitivity by Composite Regime



Note. For each of the seven asset classes, the figure plots the Pearson correlation between annualised excess returns and the core inflation shock $\varepsilon_t^{\text{core}}$, computed separately in composite-demand months (blue, $\text{comp_}R_t$ above rolling 70th percentile) and composite-supply months (red). All 7/7 asset classes show a higher absolute correlation with core inflation shocks in composite-demand months, confirming that $\text{comp_}R_t$ tracks when non-energy commodity demand is strong and core inflation risk is most priced in the cross-section. This is the parallel to Figure 25's result for energy inflation and $\text{energy_}R_t$. Source: own estimation

TABLE A.1 – Robustness: Correct-Sign Count Across Rolling Windows (12 to 60 Months)

TABLE 8 – Robustness: Conditional Energy Beta Difference across Rolling Windows

$\beta_{\text{H}} = \beta_{\text{H}} - \beta_{\text{S}}$. (1) is defined for risk, all for safe. Window = rolling window for energy β and demand dummy. Baseline = 24m highlighted.

Win.	N	Stocks	Treasuries	Corp.Bonds	Currencies	Com.	REITs	Intl.Stks	4/7	Sig/7
		(-)	(-)	(-)	(+)	(-)	(-)	(+)		
12m	244	+0.19 (+0.7)	-0.22** (-2.1)	+0.27*** (+2.5)	+0.01 (+0.3)	-0.03 (-0.1)	+0.21 (+0.8)	+0.28 (+1.0)	4/7	2/7
18m	244	+0.24 (+0.9)	-0.10 (-0.3)	-0.21*** (-2.0)	+0.07 (+0.7)	+0.06 (+0.2)	+0.11 (+0.3)	+0.11 (+0.4)	7/7	1/7
24m	244	+0.18 (+0.7)	-0.20** (-1.9)	-0.09 (-1.3)	+0.09 (+1.0)	+0.07** (+2.0)	+0.29 (+0.9)	+0.11 (+0.3)	7/7	2/7
36m	232	+0.32** (+1.8)	-0.20** (-1.8)	+0.80 (+3.0)	-0.00 (-0.0)	+0.09 (+0.4)	+0.39*** (+2.1)	+0.16 (+0.5)	5/7	3/7
48m	220	+0.27 (+1.0)	-0.07 (-0.7)	+0.80 (+3.7)	+0.01 (+0.0)	-0.03 (-0.1)	+0.81*** (+2.8)	+0.28 (+1.1)	5/7	1/7
60m	206	+0.13 (+0.5)	-0.02 (-0.2)	+0.00 (+0.0)	-0.01 (-0.4)	-0.24 (-1.1)	+0.70*** (+2.0)	+0.11 (+0.4)	4/7	1/7

KEY: 4-7/7 correct signs across all windows 12m-60m. Result is robust to window choice. Baseline 24m highlighted.

(Notes: DFF (upper) and t-stat in parentheses (lower) per cell. H3=correct sign; H3=incorrect. *** p<0.01, ** p<0.05, * p<0.10. Respective signs (+) = DFF>0 for risk assets; (-) = DFF<0 for safe assets.

Note. The table repeats the first-pass conditional beta estimation for rolling windows of 12, 18, 24, 36, 48, and 60 months used to construct $\text{energy_}R_t$ and the demand dummy. “Correct” = number of asset classes satisfying H3 ($\beta_D > \beta_S$ for risk assets, $\beta_D < \beta_S$ for safe assets). “Sig” = number with $|t| > 1.645$. The 24-month baseline (row highlighted) produces 7/7 correct signs and 2 significant differences. Correct signs range from 4/7 (12-month, 60-month windows) to 7/7 (18- and 24-month windows), confirming that the result is robust to window choice.

Table A.2 — Cross-Indicator Comparison: energy_ R_t versus comp_ R_t Left (red): energy_ R_t dummy — first-pass betas (Script 12). Right (blue): composite_ R_t dummy (non-energy commodities). H3: Diff>0 risk, <0 safe. Core-sens: |corr(r_t , $\varepsilon_t^{\text{core}}$)| higher in comp-demand months.

Class	energy β_D	energy_ R_t (Script 12) energy β_S	Diff	H3	comp β_D	composite_ R_t (Script 14) comp β_S	Diff	H3	Core-sens
Expected	>0 risk <0 safe	=0	>0/>	7/7	>0 risk <0 safe	=0	>0/<0	5/7	7/7
Stocks	+0.256	+0.117	+0.140	YES	+0.255	+0.138	+0.117	YES	YES
Treasuries	-0.285	-0.065	-0.220	YES	-0.241	-0.113	-0.128	YES	YES
Corp.Bonds	-0.129	-0.025	-0.104	YES	-0.142	-0.035	-0.107	YES	YES
Currencies	+0.206	+0.098	+0.108	YES	+0.113	+0.159	-0.046	NO	YES
Commodities	+0.714	+0.204	+0.510	YES	+0.655	+0.299	+0.356	YES	YES
REITs	+0.386	+0.111	+0.275	YES	+0.670	+0.049	+0.621	YES	YES
Intl.Stocks	+0.312	+0.206	+0.106	YES	+0.229	+0.251	-0.022	NO	YES

energy_ R_t : 7/7 correct energy beta signs (Script 12). composite_ R_t : 5/7 correct energy betas, 7/7 correct CORE sensitivity. Each indicator does a different job: energy_ R_t -> energy risk; composite_ R_t -> core risk.Notes: Beta columns: first-pass OLS with regime-interacted energy shock. H3: Diff = beta_D - beta_S, expected >0 for risk assets. Core-sens: |corr(r_t , $\varepsilon_t^{\text{core}}$)| higher in composite-demand months. *** p<0.01 ** p<0.05 * p<0.10. NW-HAC (bw=12). Mar 2003 - Jun 2023.

Notes. Columns 2 to 5 reproduce the energy beta results from Table 7 ($\hat{\beta}^D$, $\hat{\beta}^S$, Diff = $\hat{\beta}^D - \hat{\beta}^S$, H3 correct sign). Columns 6 to 10 report parallel first-pass estimates using the comp_ R_t demand dummy instead of the energy_ R_t dummy, and the core sensitivity test: whether |corr(r_t , $\varepsilon_t^{\text{core}}$)| is larger in composite-demand months than in composite-supply months. energy_ R_t produces 7/7 correct H3 signs on energy betas. comp_ R_t produces 5/7 correct H3 signs on energy betas but 7/7 correct core sensitivity signs. The two indicators carry distinct information: energy_ R_t identifies when energy inflation risk is priced, comp_ R_t identifies when core inflation risk is most priced. corr(energy_ R_t , comp_ R_t) = +0.28. $N = 244$ (219 for Currencies).